

第一章

1. 验证函数 $y = cx^3$ (c 是常数) 是方程 $3y - xy' = 0$ 的解。

证明: $y' = 3cx^2 \implies 3cx^3 - x3cx^2 = 0$ 。

2. 验证函数 $y = cx + \frac{1}{c}$ (c 是常数) 和 $y = \pm 2\sqrt{x}$ 都是方程 $y = xy' + \frac{1}{y'}$ 的解。

证明: $y = cx + \frac{1}{c}$, $y' = c \implies xy' + \frac{1}{y'} = cx + \frac{1}{c} = y$ 。

$y = \pm 2\sqrt{x}$, $y' = \pm \frac{1}{\sqrt{x}} \implies xy' + \frac{1}{y'} = \pm 2\sqrt{x} = y$ 。

3. 验证参数变量方程 $x = t^3 - t + 2$, $y = \frac{3}{4}t^4 - \frac{1}{2}t^2 + c$ (c 是常数, t 是参变量) 所决定的函数 y 满足方程 $x = (\frac{dy}{dx})^2 - \frac{dy}{dx} + 2$ 。

证明: $\frac{dx}{dt} = 3t^2 - 1$, $\frac{dy}{dt} = 3t^3 - t \implies \frac{dy}{dx} = \frac{3t^3 - t}{3t^2 - 1} = t$
 $\implies \frac{dy^3}{dx} - \frac{dy}{dx} + 2 = t^3 - t + 2 = x$ 。

4. 验证函数 $y = c_1 \cos kx + c_2 \sin kx$ (k, c_1, c_2 是常数) 是方程 $y'' + k^2 y = 0$ 的解。

证明: $y = c_1 \cos kx + c_2 \sin kx \implies y' = -c_1 k \sin kx + c_2 k \cos kx \implies y'' = -c_1 k^2 \cos kx - c_2 k^2 \sin kx \implies y'' + k^2 y = -c_1 k^2 \cos kx - c_2 k^2 \sin kx + c_1 k^2 \cos kx + c_2 k^2 \sin kx$ 。

5. 验证函数 $y = -6 \cos 2x + 8 \sin 2x$ 是方程的 $y'' + y' + \frac{5}{2}y = 25 \cos 2x$ 解, 且满足初值条件 $y(0) = -6$, $y'(0) = 16$ 。

证明: $y = -6 \cos 2x + 8 \sin 2x \implies y' = 12 \sin 2x + 16 \cos 2x \implies y'' = 24 \cos 2x - 32 \sin 2x \implies y'' + y' + \frac{5}{2}y = 24 \cos 2x - 32 \sin 2x + 12 \sin 2x + 16 \cos 2x - 15 \cos 2x + 20 \sin 2x = 25 \cos 2x$, 且 $y(0) = -6$, $y'(0) = 16$ 。

求下列可分离变量方程的解 (6-10):

6. $\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0$ 。

解: $\frac{dx}{\sqrt{1-x^2}} = -\frac{ydy}{\sqrt{1-y^2}} \implies \arcsin x + c = \sqrt{1-y^2}$, 及 $y = \pm 1$ 。

7. $y' = (2y+1) \cot x$, $y(\frac{\pi}{4}) = \frac{1}{2}$ 。

解: $\frac{dy}{2y+1} = \frac{\cos x dx}{\sin x} \implies \frac{1}{2} \ln |2y+1| + c = \ln |\sin x| \implies \sqrt{|2y+1|} = c \sin x \implies 2y+1 = c \sin^2 x$, $y = \frac{c}{2} \sin^2 x - \frac{1}{2} \implies y(\frac{\pi}{4}) = \frac{1}{4}c - \frac{1}{2} = \frac{1}{2}$,

$$c = 4 \implies y = 4 \sin^2 x - \frac{1}{2}$$

$$8. y' = 2\sqrt{y} \ln x, \quad y(e) = 1.$$

$$\text{解: } \frac{dy}{2\sqrt{y}} = \ln x dx \implies \sqrt{y} = x \ln x - x + c, \text{ 而 } \sqrt{y(e)} = 1 = e - e + c \implies$$

$$c = 1 \implies y = (x \ln x - x + 1)^2, \quad y \equiv 0 \text{ 时不合理} \implies y = (x \ln x - x + 1)^2.$$

$$9. 2(x^2 - 1)yy' = (2x + 3)(1 + y^2).$$

$$\text{解: } \frac{ydy}{1 + y^2} = \frac{(2x + 3)dx}{2(x^2 - 1)} \implies \frac{1}{2} \ln |1 + y^2| = \frac{5}{4} \ln |x - 1| - \frac{1}{4} \ln |x + 1| + c \implies$$

$$\sqrt{1 + y^2} = c \left(\frac{x-1}{x+1} \right)^{\frac{1}{4}} (x-1) \implies 1 + y^2 = c \left(\frac{x-1}{x+1} \right)^{\frac{1}{2}} (x-1)^2 \implies y^2 = c(x-1)^2 \sqrt{\frac{x-1}{x+1}} - 1.$$

$$10. y' = (1 - y^2) \tan x, \quad y(0) = 2.$$

$$\text{解: } \frac{dy}{1 - y^2} = \frac{\sin x}{\cos x} dx \implies \frac{1}{2} \ln \left| \frac{1+y}{1-y} \right| = -\ln |\cos x| + c \implies \sqrt{\frac{1+y}{1-y}} =$$

$$c \frac{1}{\cos x} \implies \frac{1-y}{1+y} = c \cos^2 x \implies y = \frac{1 - c \cos^2 x}{1 + c \cos^2 x}, \quad y(0) = \frac{1-c}{1+c} = 2 \implies c = -\frac{1}{3} \implies y = \frac{3 + \cos^2 x}{3 - \cos^2 x}.$$

求下列齐次方程的解 (11-17) :

$$11. \frac{dy}{dx} = \frac{2xy}{x^2 + y^2}.$$

$$\text{解: 令 } y = ux, \quad \frac{xdy + udx}{dx} = \frac{2x^2u}{x^2 + u^2x^2} = \frac{2u}{1 + u^2} \implies \frac{xdu}{dx} + u = \frac{2u}{1 + u^2} \implies$$

$$x \frac{du}{dx} = \frac{u - u^3}{1 + u^2} \implies \frac{1 + u^2}{u - u^3} du = \frac{1}{x} dx \implies \ln |u| - \ln |1 - u| - \ln |1 + u| =$$

$$\ln |x| + c \implies \frac{u}{(1-u)(1+u)} = cx \implies \frac{u}{1-u^2} = cx \implies \frac{\frac{y}{x}}{1 - \frac{y^2}{x^2}} = cx \implies$$

$$\frac{xy}{x^2 - y^2} = cx \implies \frac{y}{x^2 - y^2} = c, \text{ 或 } y = \pm x.$$

$$12. \frac{dy}{dx} = \frac{y}{x} (1 + \ln y - \ln x).$$

$$\text{解: } \frac{dy}{dx} = \frac{y}{x} \left(1 + \ln \frac{y}{x} \right), \text{ 令 } y = ux \implies x \frac{du}{dx} + u = u(1 + \ln u) \implies x \frac{du}{dx} =$$

$$u \ln u \implies \frac{1}{u \ln u} du = \frac{1}{x} dx \implies \ln |\ln |u|| = \ln |x| + c \implies \ln |u| = cx, \quad u = e^{cx},$$

$$x > 0 \implies y = xe^{cx}.$$

$$13. y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}.$$

$$\text{解: } \frac{y^2}{x^2} + \frac{ydy}{x dx}, \text{ 令 } xu = y \implies u^2 + x \frac{du}{dx} + u = u \left(x \frac{du}{dx} + u \right) \implies u =$$

$$(u-1)x \frac{du}{dx} \Rightarrow \frac{dx}{x} = \frac{u-1}{u} du \Rightarrow u - \ln|u| = \ln|x| + c \Rightarrow \frac{e^u}{u} = cx \Rightarrow e^{\frac{y}{x}} = cy \Rightarrow y = ce^{\frac{y}{x}}.$$

$$14. (y+x)dy = (y-x)dx.$$

解: $\frac{dy}{dx} = \frac{y-x}{y+x} = \frac{\frac{y}{x}-1}{\frac{y}{x}+1}$, 令 $y = ux \Rightarrow x \frac{du}{dx} + u = \frac{u-1}{u+1} \Rightarrow x \frac{du}{dx} = -\frac{1+u^2}{1+u} \Rightarrow \frac{1+u}{1+u^2} du = -\frac{dx}{x} \Rightarrow \arctan u + \frac{1}{2} \ln(1+u^2) = -\ln|x| + c \Rightarrow e^{\arctan u} \sqrt{1+u^2} = \frac{c}{|x|} \Rightarrow e^{\arctan u} \sqrt{x^2+y^2} = c \Rightarrow c\sqrt{x^2+y^2} = e^{-\arctan \frac{y}{x}}.$

$$15. (x-y \cos \frac{y}{x})dx + x \cos \frac{y}{x} dy = 0.$$

解: $(1-\frac{y}{x} \cos \frac{y}{x})dx + \cos \frac{y}{x} dy = 0$, 令 $y = ux \Rightarrow (1-u \cos u) + \cos u (x \frac{du}{dx} + u) = 0 \Rightarrow 1 + x \cos u \frac{du}{dx} = 0 \Rightarrow \cos u du = -\frac{dx}{x}$, $\sin u = -\ln|x| + c \Rightarrow e^{\sin u} = \frac{c}{x} \Rightarrow x e^{\sin \frac{y}{x}} = c \Rightarrow \sin \frac{y}{x} = \ln \frac{c}{x}.$

$$16. \frac{dy}{dx} = 2\sqrt{\frac{y}{x}} + \frac{y}{x}, \quad y(1) = 4.$$

解: 令 $y = ux$, $u \geq 0 \Rightarrow x \frac{du}{dx} + u = 2\sqrt{u} + u \Rightarrow \frac{1}{2\sqrt{u}} = \frac{1}{x} dx \Rightarrow \sqrt{u} = \ln|x| + c \Rightarrow e^{\sqrt{u}} = cx \Rightarrow e^{\sqrt{\frac{y}{x}}} = cx$, $y(1) = 4 \Rightarrow e^{\sqrt{\frac{4}{1}}} = c$, $c = e^2 \Rightarrow e^{\sqrt{\frac{y}{x}}} = e^2 x \Rightarrow \sqrt{\frac{y}{x}} = 2 + \ln x.$

$$17. xy' - y = \sqrt{x^2 - y^2}, \quad y(1) = \frac{1}{2}.$$

解: 令 $y = ux$, $x(x \frac{du}{dx} + u) - ux = \sqrt{x^2 - x^2 u^2} \Rightarrow x^2 \frac{du}{dx} = |x| \sqrt{1-u^2} \Rightarrow$
若 $x > 0$, $\frac{du}{\sqrt{1-u^2}} = \frac{dx}{x} \Rightarrow \arcsin u = \ln x + c$; 若 $x < 0$, $\frac{du}{\sqrt{1-u^2}} = -\frac{dx}{x} \Rightarrow \arcsin u = -\ln -x + c$. $y(1) = \frac{1}{2} \Rightarrow \frac{1}{2} = u \times 1 = u \Rightarrow x > 0 \Rightarrow \arcsin \frac{1}{2} = \ln 1 + c$, $c = \frac{\pi}{6} \Rightarrow \arcsin \frac{y}{x} = \ln x + \frac{\pi}{6}.$

求下列一阶线性方程或伯努利方程的解 (18-24):

$$18. \frac{dy}{dx} = x^2 - \frac{y}{x}.$$

解: $y' + \frac{1}{x}y = x^2$, $p(x) = \frac{1}{x}$, $f(x) = x^2$, $e^{-\int p(x)dx} = e^{-\int \frac{1}{x}dx} = \frac{1}{x} \Rightarrow y = \frac{1}{x}(\int x^3 dx + c) = \frac{1}{x}(\frac{1}{4}x^4 + c) = \frac{1}{4}x^3 + \frac{c}{x}.$

$$19. xy' - y = x^3 e^{-x}.$$

解: $\frac{dy}{dx} - \frac{1}{x}y = x^2e^{-x}$, $p(x) = -\frac{1}{x}$, $f(x) = x^2e^{-x}$, $e^{-\int p(x)dx} = x$,
 $\Rightarrow y = x(\int x^2e^{-x}\frac{1}{x}dx + c) = x(\int xe^{-x}dx + c) = x(-xe^{-x} - e^{-x} + c)$.

20. $\frac{dy}{dx} + 2xy + x = e^{-x^2}$, $y(0) = 2$.

解: $\frac{dy}{dx} + 2xy = e^{-x^2} - x$, $p(x) = 2x$, $f(x) = e^{-x^2} - x$, $e^{-\int p(x)dx} = e^{-x^2} \Rightarrow$
 $y = e^{-x^2}(\int (e^{-x^2} - x)e^{x^2}dx + c) = e^{-x^2}(\int (1 - xe^{x^2})dx + c) = e^{-x^2}(x - \frac{1}{2}e^{x^2} +$
 $c) = (c + x)e^{-x^2} - \frac{1}{2}$, $y(0) = c - \frac{1}{2} = 2 \Rightarrow c = \frac{5}{2} \Rightarrow y = (\frac{5}{2} + x)e^{-x^2} - \frac{1}{2}$.

21. $xy' = x \cos x - 2 \sin x - 2y$, $y(\pi) = 0$.

解: $\frac{dy}{dx} + \frac{2}{x}y = \cos x - \frac{2}{x} \sin x$, $p(x) = \frac{2}{x}$, $f(x) = \cos x - \frac{2}{x} \sin x \Rightarrow$
 $e^{-\int p(x)dx} = (\frac{1}{x})^2$, $y = \frac{1}{x^2}(\int (\cos x - \frac{2}{x} \sin x)x^2dx + c)$
 $= \frac{1}{x^2}(\int (x^2 \cos x - 2x \sin x)dx + c)$
 $= \frac{1}{x^2}(x^2 \sin x + 2x \cos x - 2 \sin x + 2x \cos x - 2 \sin x + c)$
 $= \frac{1}{x^2}(x^2 \sin x + 4x \cos x - 4 \sin x + c)$.
 $y(\pi) = \frac{1}{\pi^2}(4\pi + c) = 0 \Rightarrow c = -4\pi \Rightarrow y = \frac{1}{x^2}(x^2 - 4) \sin x + \frac{4}{x} \cos x - \frac{4\pi}{x^2} =$
 $(1 - \frac{4}{x^2}) \sin x + \frac{4}{x} \cos x - \frac{4\pi}{x^2}$.

22. $\frac{dy}{dx} - \frac{xy}{2(x^2-1)} - \frac{x}{2y} = 0$, $y(0) = 1$.

解: 两边乘以 y , $y \frac{dy}{dx} - \frac{xy^2}{2(x^2-1)} - \frac{x}{2} = 0$, 令 $z = y^2 \Rightarrow \frac{1}{2} \frac{dz}{dx} - \frac{xz}{2(x^2-1)} =$
 $\frac{x}{2} \Rightarrow \frac{dz}{dx} - \frac{xz}{x^2-1} = x$.
 $p(x) = -\frac{x}{x^2-1}$, $f(x) = x \Rightarrow e^{-\int p(x)dx} = e^{\int \frac{x}{x^2-1}dx} = \sqrt{|x^2-1|}$, 这里初
 值是 $x=0$ 取 $x^2 < 1$.
 $z = \sqrt{1-x^2}(\int \frac{x}{\sqrt{1-x^2}}dx + c) = \sqrt{1-x^2}(-\sqrt{1-x^2} + c)$, $y^2 = (x^2 - 1 +$
 $c\sqrt{1-x^2})$.
 $y(0) = 1 > 0 \Rightarrow y = \sqrt{x^2 - 1 + c\sqrt{1-x^2}}$.
 $y(0) = \sqrt{-1 + c} = 1 \Rightarrow -1 + c = 1 \Rightarrow c = 2 \Rightarrow y = \sqrt{x^2 - 1 + 2\sqrt{1-x^2}}$.

23. $xy' - 4y = x^2 \sqrt{y}$.

解: $\frac{dy}{dx} - \frac{4}{x}y = x\sqrt{y}$, 两边同除以 $\sqrt{y}$, 令 $z = \sqrt{y} \Rightarrow 2\frac{dz}{dx} - \frac{4}{x}z = x \Rightarrow$
 $\frac{dz}{dx} - \frac{2}{x}z = \frac{x}{2}$.

$$p(x) = -\frac{2}{x}, \quad f(x) = \frac{x}{2}.$$

$$e^{-\int p(x)dx} = e^{\int \frac{2}{x}dx} = x^2 \implies z = x^2 \left(\int \frac{x}{2x^2} dx + c \right) = x^2 \left(\frac{1}{2} \ln|x| + c \right) \implies$$

$$\sqrt{y} = \frac{x^2}{2} \ln|x| + cx^2 \text{ 或 } y = 0.$$

$$24. \frac{dy}{dx} = \frac{y^2 - x}{2xy}.$$

解: $\frac{2ydy}{dx} = \frac{y^2 - x}{x}, \text{ 令 } z = y^2 \implies \frac{dz}{dx} = \frac{z}{x} - 1 \implies \frac{dz}{dx} - \frac{1}{x}z = -1.$

$$p(x) = -\frac{1}{x}, \quad f(x) = -1, \quad e^{-\int p(x)dx} = x \implies z = x \left(\int -1 \times \frac{1}{x} dx + c \right) =$$

$$x(-\ln|x| + c) \implies y^2 = cx - x \ln|x|.$$

验证下列方程为全微分方程或找出积分因子, 然后求其解 (25-36):

$$25. (5x^4 y dx + x^5 dy) + x^3 dx = 0.$$

解: $(5x^4 y + x^3) dx + x^5 dy = 0 \implies \frac{\partial(5x^4 y + x^3)}{\partial y} = 5x^4, \quad \frac{\partial x^5}{\partial x} = 5x^4 \implies$ 是全微分方程,

$$u(x, y) = \int_{x_0}^x (5x^4 y_0 + x^3) dx + \int_{y_0}^y x^5 dy$$

$$= x^5 y_0 - x_0^5 y_0 + \frac{1}{4} x^4 - \frac{1}{4} x_0^4 + x^5 y - x^5 y_0$$

$$= x^5 y + \frac{1}{4} x^4 - x_0^5 - \frac{1}{4} x_0^4 = c$$

$$\implies x^5 y + \frac{1}{4} x^4 = c.$$

$$26. 2(y dx + x dy) + x dx - 5y dy = 0, \quad y(0) = 1.$$

解: $(2y + x) dx + (2x - 5y) dy = 0 \implies \frac{\partial(2y + x)}{\partial y} = 2, \quad \frac{\partial(2x - 5y)}{\partial x} = 2 \implies$ 是全微分方程,

$$u(x, y) = \int_{x_0}^x (2y_0 + x) dx + \int_{y_0}^y (2x - 5y) dy$$

$$= 2y_0 x - 2x_0 y_0 + \frac{1}{2} x^2 - \frac{1}{2} x_0^2 + 2xy - 2xy_0 - \frac{5}{2} y^2 + \frac{5}{2} y_0^2$$

$$= \frac{1}{2} x^2 - \frac{5}{2} y^2 + 2xy - 2x_0 y_0 - \frac{1}{2} x_0^2 + \frac{5}{2} y_0^2 = c$$

$$\implies \frac{1}{2} x^2 - \frac{5}{2} y^2 + 2xy = c, \quad y(0) = 1 \implies c = -\frac{5}{2} \implies \frac{1}{2} x^2 - \frac{5}{2} y^2 + 2xy + \frac{5}{2} =$$

$$0 \implies x^2 - 5y^2 + 4xy + 5 = 0.$$

$$27. \frac{xdx + ydy}{\sqrt{1+x^2+y^2}} + \frac{ydx - xdy}{\sqrt{x^2+y^2}} = 0.$$

解: $\left(\frac{x}{\sqrt{1+x^2+y^2}} + \frac{y}{x^2+y^2} \right) dx + \left(\frac{y}{\sqrt{1+x^2+y^2}} - \frac{x}{x^2+y^2} \right) dy,$

$$\frac{\partial(\frac{x}{\sqrt{1+x^2+y^2}} + \frac{y}{x^2+y^2})}{\partial y} = \frac{xy}{(1+x^2+y^2)^{\frac{3}{2}}} + \frac{x^2-y^2}{(x^2+y^2)^2},$$

$$\frac{\partial(\frac{y}{\sqrt{1+x^2+y^2}} - \frac{x}{x^2+y^2})}{\partial x} = \frac{xy}{(1+x^2+y^2)^{\frac{3}{2}}} - \frac{y^2-x^2}{(x^2+y^2)^2} = \frac{xy}{(1+x^2+y^2)^{\frac{3}{2}}} + \frac{x^2-y^2}{(x^2+y^2)^2},$$

$\Rightarrow$ 是全微分方程,

$$\begin{aligned} u(x, y) &= \int_{x_0}^x (\frac{x}{\sqrt{1+x^2+y_0^2}} + \frac{y_0}{x^2+y_0^2}) dx + \int_{y_0}^y (\frac{y}{\sqrt{1+x^2+y^2}} - \frac{x}{x^2+y^2}) dy \\ &= \sqrt{1+x^2+y_0^2} - \sqrt{1+x_0^2+y_0^2} + \arctan \frac{x}{y_0} - \arctan \frac{x_0}{y_0} \\ &\quad + \sqrt{1+x^2+y^2} - \sqrt{1+x^2+y_0^2} - \arctan \frac{y}{x} + \arctan \frac{y_0}{x} \\ &= c \\ \Rightarrow \sqrt{1+x^2+y_0^2} &= \arctan \frac{y}{x} + c. \end{aligned}$$

$$28. (ye^x - e^{-y})dx + (xe^{-y} + e^x)dy = 0.$$

解: $\frac{\partial(ye^x - e^{-y})}{\partial y} = e^x + e^{-y}, \quad \frac{\partial(xe^{-y} + e^x)}{\partial x} = e^{-y} + e^x \Rightarrow$ 是全微分方程,

$$\begin{aligned} u(x, y) &= \int_{x_0}^x (y_0 e^x - e^{-y_0}) dx + \int_{y_0}^y (xe^{-y} + e^x) dy \\ &= y_0 e^x - y_0 e^{x_0} - xe^{y_0} + x_0 e^{y_0} - xe^{-y} + xe^{-y_0} + ye^x - y_0 e^x = c \\ \Rightarrow ye^x - xe^{-y} &= c. \end{aligned}$$

$$29. (\frac{1}{x} - \frac{y^2}{(x-y)^2})dx + (\frac{x^2}{(x-y)^2} - \frac{1}{y})dy = 0.$$

解: $\frac{\partial(\frac{1}{x} - \frac{y^2}{(x-y)^2})}{\partial y} = -\frac{2xy}{(x-y)^3}, \quad \frac{\partial(\frac{x^2}{(x-y)^2} - \frac{1}{y})}{\partial x} = -\frac{2xy}{(x-y)^3} \Rightarrow$ 是全微分方程,

$$\begin{aligned} u(x, y) &= \int_{x_0}^x (\frac{1}{x} - \frac{y_0^2}{(x-y_0)^2}) dx + \int_{y_0}^y (\frac{x^2}{(x-y)^2} - \frac{1}{y}) dy \\ &= \ln|x| - \ln|x_0| + y_0^2 \frac{1}{x-y_0} - y_0^2 \frac{1}{(x_0-y_0)} + \frac{x^2}{x-y} - \frac{x^2}{x-y_0} \\ &\quad - \ln|y| + \ln|y_0| \\ \text{而 } \frac{y_0^2}{x-y_0} - \frac{x^2}{x-y_0} &= -\frac{(x-y_0)(x+y_0)}{x-y_0} = -x-y_0 \\ \Rightarrow \ln|x| - \ln|y| + \frac{x^2}{x-y} - x &= \ln|x| - \ln|y| + \frac{xy}{x-y} = c. \end{aligned}$$

$$30. (4ydx + xdy) - x^2dx = 0.$$

解: $(4y - x^2)dx + xdy = 0, \quad \frac{\partial(4y - x^2)}{\partial y} = 4, \quad \frac{\partial x}{\partial x} = 1 \Rightarrow$ 不是全微分方程.

$$\begin{aligned} \varphi(x) &= \frac{4-1}{x} = \frac{3}{x}, \quad \mu = e^{\int \frac{3}{x} dx} = x^3 \\ \Rightarrow (4x^3y - x^5)dx + x^4dy &= 0 \end{aligned}$$

$$\begin{aligned}
\Rightarrow u(x, y) &= \int_{x_0}^x (4x^3 y_0 - x^5) dx + \int_{y_0}^y x^4 dy \\
&= x^4 y_0 - x_0^4 y_0 - \frac{1}{6} x^6 + \frac{1}{6} x_0^6 + x^4 y - x^4 y_0 \\
\Rightarrow x^4 y - \frac{1}{6} x^6 &= c .
\end{aligned}$$

$$\begin{aligned}
&31. (2xy dx - 3x^2 dy) + y^2 dy = 0 . \\
\text{解: } &2xy dx + (y^2 - 3x^2) dy , \quad \frac{\partial(2xy)}{\partial y} = 2x , \quad \frac{\partial(y^2 - 3x^2)}{\partial x} = -6x \\
\Rightarrow \psi(y) &= \frac{2x + 6x}{-2xy} = -\frac{4}{y} , \quad \mu = e^{\int -\frac{4}{y} dy} = \frac{1}{y^4} \\
\Rightarrow &2xy^{-3} dx + (y^{-2} - 3x^2 y^{-4}) dy = 0 \\
\Rightarrow u(x, y) &= \int_{x_0}^x 2xy_0^{-3} dx + \int_{y_0}^y (y^{-2} - 3x^2 y^{-4}) dy \\
&= x^2 y_0^{-3} - x_0^2 y_0^{-3} - y^{-1} + y_0^{-1} + x^2 y^{-3} - x^2 y_0^{-3} \\
\Rightarrow \frac{x^2}{y^3} - \frac{1}{y} &= c .
\end{aligned}$$

$$\begin{aligned}
&32. (y dx - x dy) + x^4 dx = 0 . \\
\text{解: } &(y + x^4) dx - x dy = 0 \Rightarrow \frac{\partial(y + x^4)}{\partial y} = 1 , \quad \frac{\partial - x}{\partial x} = -1 \\
\Rightarrow \varphi(x) &= \frac{1 + 1}{-x} = -\frac{2}{x} , \quad \mu = e^{-\int \frac{2}{x} dx} = \frac{1}{x^2} . \\
\Rightarrow &(yx^{-2} + x^2) dx - x^{-1} dy = 0 \\
\Rightarrow u(x, y) &= \int_{x_0}^x (y_0 x^{-2} + x^2) dx + \int_{y_0}^y -x^{-1} dy \\
&= -\frac{y_0}{x} + \frac{y_0}{x_0} + \frac{1}{3} x^3 - \frac{1}{3} x_0^3 - \frac{y}{x} + \frac{y_0}{x} \\
\Rightarrow \frac{1}{3} x^3 - \frac{y}{x} &= c .
\end{aligned}$$

$$\begin{aligned}
&33. (2xy^2 - y) dx + (2x - x^2 y) dy = 0 . \\
\text{解: } &\frac{\partial(2xy^2 - y)}{\partial y} = 4xy - 1 , \quad \frac{\partial(2x - x^2 y)}{\partial x} = 2 - 2xy \\
\Rightarrow \psi(y) &= \frac{4xy - 1 - 2 + 2xy}{-(2xy^2 - y)} = \frac{6xy - 3}{-y(2xy - 1)} = \frac{3(2xy - 1)}{-y(2xy - 1)} = -\frac{3}{y} , \\
\mu &= e^{-\int \frac{3}{y} dy} = \frac{1}{y^3} \\
\Rightarrow &(2xy^{-1} - y^{-2}) dx + (2xy^{-3} - x^2 y^{-2}) dy = 0 \\
\Rightarrow u(x, y) &= \int_{x_0}^x (2xy_0^{-1} - y_0^{-2}) dx + \int_{y_0}^y (2xy^{-3} - x^2 y^{-2}) dy \\
&= x^2 y_0^{-1} - x_0^2 y_0^{-1} - xy_0^{-2} + x_0 y_0^{-2} - xy_0^{-2} + xy_0^{-2} + x^2 y^{-1} - x^2 y_0^{-1} \\
\Rightarrow \frac{x^2}{y} - \frac{x}{y^2} &= c .
\end{aligned}$$

$$\begin{aligned}
&34. 2dx + (2x - 3y - 3) dy = 0 , \quad y(2) = 0 . \\
\text{解: } &\frac{\partial 2}{\partial y} = 0 , \quad \frac{\partial(2x - 3y - 3)}{\partial x} = 2
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \psi(y) = \frac{0-2}{-2}, \quad \mu = e^{\int 1dy} = e^y \\
&\Rightarrow 2e^y dx + (2xe^y - 3ye^y - 3e^y)dy = 0 \\
&\Rightarrow u(x, y) = \int_{x_0}^x 2e^{y_0} dx + \int_{y_0}^y (2xe^y - 3ye^y - 3e^y)dy \\
&\quad = 2e^{y_0}x - 2e^{y_0}x_0 + 2e^y x - 2e^{y_0}x - 3e^y y + 3e^{y_0}y_0 + 3e^y - 3e^{y_0} - 3e^y + 3e^{y_0} \\
&\Rightarrow 2xe^y - 3ye^y = c. \\
&y(2) = 0 \Rightarrow 4 - 0 = c \Rightarrow c = 4 \Rightarrow 2xe^y - 3ye^y = 4.
\end{aligned}$$

$$\begin{aligned}
&35. (3x^2 + 2xy - y^2)dx + (x^2 - 2xy)dy = 0. \\
\text{解: } &\frac{\partial(3x^2 + 2xy - y^2)}{\partial y} = 2x - 2y, \quad \frac{\partial(x^2 - 2xy)}{\partial x} = 2x - 2y \Rightarrow \text{是全微分方程} \\
&\Rightarrow u(x, y) = \int_{x_0}^x (3x^2 + 2xy_0 - y_0^2)dx + \int_{y_0}^y (x^2 - 2xy)dy \\
&\quad = x^3 - x_0^3 + x^2 y_0 - x_0^2 y_0 - xy_0^2 + x_0 y_0^2 + x^2 y - x^2 y_0 - xy^2 + xy_0^2 \\
&\Rightarrow x^3 + x^2 y - xy^2 = c.
\end{aligned}$$

$$\begin{aligned}
&36. (3xy^2 + 2y)dx + (2x^2 y + x)dy = 0. \\
\text{解: } &\frac{\partial(3xy^2 + 2y)}{\partial y} = 6xy + 2, \quad \frac{\partial(2x^2 y + x)}{\partial x} = 4xy + 1 \\
&\Rightarrow \varphi(x) = \frac{6xy + 2 - 4xy - 1}{2x^2 y + x} = \frac{2xy + 1}{x(2xy + 1)} = \frac{1}{x}, \quad \mu = e^{\int \frac{1}{x} dx} = x \\
&\Rightarrow u(x, y) = \int_{x_0}^x (3x^2 y_0^2 + 2xy_0)dx + \int_{y_0}^y (2x^3 y + x^2)dy \\
&\quad = x^3 y_0^2 - x_0^3 y_0^2 + x^2 y_0 - x_0^2 y_0 + x^3 y^2 - x^3 y_0^2 + x^2 y - x^2 y_0 \\
&\Rightarrow x^3 y^2 + x^2 y = c.
\end{aligned}$$

判别下列各方程的类型，并选择一种方法求解 (37-49)：

$$\begin{aligned}
&37. xy(y - xy') = x + yy', \quad y(0) = \frac{\sqrt{2}}{2}. \\
\text{解: } &xy^2 - x^2 y \frac{dy}{dx} = x + y \frac{dy}{dx} \Rightarrow y(1+x^2) \frac{dy}{dx} - xy^2 = -x \Rightarrow y \frac{dy}{dx} - \frac{x}{1+x^2} y^2 = \\
&\frac{-x}{1+x^2} \Rightarrow \frac{1}{2} \frac{dy^2}{dx} - \frac{x}{1+x^2} y^2 = -\frac{x}{1+x^2}. \\
&\text{令 } z = y^2, \quad \frac{dz}{dx} - \frac{2x}{1+x^2} z = -\frac{2x}{1+x^2}, \quad p(x) = -\frac{2x}{1+x^2}, \quad f(x) = -\frac{2x}{1+x^2}, \\
&e^{\int \frac{2x}{1+x^2} dx} = 1+x^2 \\
&\Rightarrow z = (1+x^2) \left(-\int \frac{2x}{(1+x^2)^2} + c \right) = (1+x^2) \left(\frac{1}{1+x^2} + c \right) = 1 + c(1+x^2). \\
&y(0) = \frac{\sqrt{2}}{2} > 0 \Rightarrow y = \sqrt{1 + c(1+x^2)} \Rightarrow y(0) = \sqrt{1+c} = \frac{\sqrt{2}}{2} \Rightarrow 1+c = \\
&\frac{1}{2}, \quad c = -\frac{1}{2} \Rightarrow y = \sqrt{1 - \frac{1}{2}(1+x^2)} = \sqrt{\frac{1}{2} - \frac{1}{2}x^2}.
\end{aligned}$$

$$38. \tan t \frac{dx}{dt} - x = 5 .$$

解: $\tan t \frac{dx}{dt} = 5 + x \Rightarrow \frac{dx}{5+x} = \frac{\cos t}{\sin t} dt \Rightarrow \ln |5+x| = \ln |\sin t| + c \Rightarrow$
 $5+x = c \sin t \Rightarrow x = c \sin t - 5 .$

$$39. d\theta + 2\theta r dr = r^3 dr .$$

解: $\frac{d\theta}{dr} + 2r\theta = r^3 , \quad p(r) = 2r , \quad f(r) = r^3 .$
 $e^{-\int p(r)dr=e^{-r^2}} \Rightarrow \theta = e^{-r^2} \left(\int r^3 e^{r^2} dr + c \right) = e^{-r^2} \left(\frac{1}{2} r^2 e^{r^2} - \frac{1}{2} e^{r^2} + c \right) =$
 $\frac{1}{2}(r^2 - 1) + ce^{-r^2} .$

$$40. e^y dx + (xe^y - 2y) dy = 0 .$$

解: $\frac{\partial e^y}{\partial y} = e^y , \quad \frac{\partial (xe^y - 2y)}{\partial x} = e^y \Rightarrow$ 是全微分方程
 $\Rightarrow u(x, y) = \int_{x_0}^x (e^{y_0}) dx + \int_{y_0}^y (xe^y - 2y) dy$
 $= e^{y_0} x - e^{y_0} x_0 + e^y x - e^{y_0} x - y^2 + y_0^2$
 $\Rightarrow xe^y - y^2 = c , \quad xe^y = c + y^2 \Rightarrow x = e^{-y}(c + y^2) .$

$$41. yy' + xy^2 = x .$$

解: $\frac{1}{2} \frac{dy^2}{dx} + xy^2 = x , \quad \text{令 } z = y^2$
 $\Rightarrow \frac{dz}{dx} + 2xz = 2x \Rightarrow p(x) = 2x , \quad f(x) = 2x \Rightarrow e^{-\int p(x)dx} = e^{-x^2} .$
 $z = e^{-x^2} \left(\int 2xe^{x^2} dx + c \right) = e^{-x^2} (e^{x^2} + c) = 1 + ce^{-x^2} \Rightarrow y^2 = 1 + ce^{-x^2} .$

$$42. xyy' = x^2 + y^2 .$$

解: $y \frac{dy}{dx} = x + \frac{y^2}{x} , \quad \frac{1}{2} \frac{dy^2}{dx} - \frac{1}{x} y^2 = x , \quad \text{令 } z = y^2 \Rightarrow \frac{dz}{dx} - \frac{2}{x} z = 2x \Rightarrow$
 $p(x) = -\frac{2}{x} , \quad f(x) = 2x \Rightarrow e^{-\int p(x)dx} = x^2 \Rightarrow z = x^2 \left(\int 2xx^{-2} dx + c \right) =$
 $x^2 \left(\int \frac{2}{x} dx + c \right) = x^2 (2 \ln |x| + c) \Rightarrow y^2 = 2x^2 \ln(cx) .$

$$43. ydx - xdy = x^2 y dy .$$

解: $\frac{dx}{dy} - \frac{x}{y} = x^2 \Rightarrow x^{-2} \frac{dx}{dy} - \frac{1}{y} \frac{1}{x} = 1 \Rightarrow -\frac{d\frac{1}{x}}{dy} - \frac{1}{y} \frac{1}{x} = 1 .$
 令 $z = \frac{1}{x}$
 $\Rightarrow \frac{dz}{dy} + \frac{1}{y} z = -1 \Rightarrow p(y) = \frac{1}{y} , \quad f(y) = -1 , \quad e^{-\int p(y)dy} = \frac{1}{y} \Rightarrow z =$
 $\frac{1}{y} \left(\int -1 y dy + c \right) = \frac{1}{y} \left(-\frac{1}{2} y^2 + c \right) = -\frac{1}{2} y + \frac{c}{y} \Rightarrow \frac{1}{x} = -\frac{1}{2} y + \frac{c}{y} \Rightarrow \frac{y}{x} =$
 $-\frac{1}{2} y^2 + c \Rightarrow \frac{y}{x} + \frac{1}{2} y^2 = c .$

$$44. (y^2 + x)dx - 2xydy = 0 .$$

$$\text{解: } \frac{\partial(y^2 + x)}{\partial y} = 2y, \quad \frac{\partial(-2xy)}{\partial x} = -2y \implies \varphi(x) = \frac{2y + 2y}{-2xy} = -\frac{2}{x},$$

$$\mu(x) = e^{\int \frac{2}{x} dx} = \frac{1}{x^2} \implies (x^{-2}y^2 + x^{-1})dx - 2x^{-1}ydy = 0 .$$

$$\begin{aligned} u(x, y) &= \int_{x_0}^x (x^{-2}y_0^2 + x^{-1})dx - 2 \int_{y_0}^y x^{-1}ydy \\ &= -x^{-1}y_0^2 + x_0^{-1}y_0^2 + \ln|x| - \ln|x_0| - x^{-1}y^2 + x^{-1}y_0^2 \end{aligned}$$

$$\implies \ln|x| - x^{-1}y^2 = c \implies y^2 = x(-\ln|x| + c) .$$

$$45. (x - y)dx + xdy = 0 .$$

$$\text{解: } \frac{dy}{dx} - \frac{y}{x} = -1, \implies p(x) = -\frac{1}{x}, \quad f(y) = -1, \implies e^{-\int p(x)dx} = x \implies$$

$$y = x\left(\int -\frac{1}{x}dx + c\right) = x(-\ln|x| + c) .$$

$$46. \frac{dy}{dx} = \frac{y}{x + y^2} .$$

$$\text{解: } \frac{dx}{dy} = \frac{x}{y} + y^2, \quad \frac{dx}{dy} - \frac{1}{y}x = y^2, \quad p(y) = -\frac{1}{y}, \quad f(y) = y^2,$$

$$e^{-\int p(y)dy} = y \implies x = y\left(\int y^2 \frac{1}{y}dy + c\right) = y\left(\frac{1}{2}y^2 + c\right) = \frac{1}{2}y^3 + cy .$$

$$47. (xy + 1)ydx - xdy = 0 .$$

$$\text{解: } y^2 + \frac{y}{x} = \frac{dy}{dx} \implies \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{x} \frac{1}{y} = 1, \quad \text{令 } z = \frac{1}{y} \implies -\frac{dz}{dx} - \frac{1}{x}z = 1 \implies$$

$$\begin{aligned} \frac{dz}{dx} + \frac{1}{x}z &= -1, \quad p(x) = \frac{1}{x}, \quad f(x) = -1 \implies e^{-\int p(x)dx} = \frac{1}{x} \implies z = \\ \frac{1}{x}\left(\int -xdx + c\right) &= \frac{1}{x}\left(-\frac{1}{2}x^2 + c\right) = -\frac{1}{2}x + \frac{c}{x} = \frac{c - x^2}{2x} = \frac{1}{y} \implies y = \frac{2x}{c - x^2} \text{ 或 } \\ y &= 0 . \end{aligned}$$

$$48. (x^2 + y^2)dy + 2xydx = 0 .$$

$$\text{解: } \frac{\partial(x^2 + y^2)}{\partial x} = 2x, \quad \frac{\partial(2xy)}{\partial y} = 2x \implies \text{是全微分方程.}$$

$$\begin{aligned} u(x, y) &= \int_{y_0}^y (x_0^2 + y^2)dy + \int_{x_0}^x 2xydy \\ &= x_0^2y - x_0^2y_0 + \frac{1}{3}y^3 - \frac{1}{3}y_0^3 + x^2y - x_0^2y \end{aligned}$$

$$\implies \frac{1}{3}y^3 + x^2y = c \text{ 及 } y = 0 \text{ (已包括于 } \frac{1}{3}y^3 + x^2y = c)$$

$$49. (y - x^2)y' + 4xy = 0 .$$

$$\text{解: } 4xydx + (y - x^2)dy = 0, \quad \frac{\partial(4xy)}{\partial y} = 4x, \quad \frac{\partial(y - x^2)}{\partial x} = -2x \implies \psi(y) =$$

$$\frac{4x + 2x}{-4xy} = -\frac{3}{2y}, \quad \mu(y) = e^{-\int \frac{3}{2y}dy} = \left(\frac{1}{y}\right)^{\frac{3}{2}} \implies 4xy^{-\frac{1}{2}}dx + (y^{-\frac{1}{2}} - x^2y^{-\frac{3}{2}})dy = 0$$

$$u(x, y) = \int_{x_0}^x 4xy_0^{-\frac{1}{2}}dx + \int_{y_0}^y (y^{-\frac{1}{2}} - x^2y^{-\frac{3}{2}})dy$$

$$= 2x^2y_0^{-\frac{1}{2}} - 2x_0^2y_0^{-\frac{1}{2}} + 2y^{\frac{1}{2}} - 2y_0^{\frac{1}{2}} + 2x^2y^{-\frac{1}{2}} - 2x_0^2y_0^{-\frac{1}{2}} \\ \Rightarrow 2y^{\frac{1}{2}} + 2x^2y^{-\frac{1}{2}} = c \Rightarrow y^{\frac{1}{2}} + x^2y^{-\frac{1}{2}} = c \Rightarrow x^2 = -y + c\sqrt{y} \text{ 或 } y = 0.$$

50. 设 $f(x)$ 是连续函数, 并且满足 $f(x) + 2 \int_0^x f(t)dt = x^2$. 求 $f(x)$.

解: $f(x) + 2 \int_0^x f(t)dt = x^2 \Rightarrow f'(x) + 2f(x) = 2x \Rightarrow p(x) = 2$,
 $e^{-2} \int dx = e^{-2x}$.
 $f(x) = e^{-2x} \left(\int 2xe^{2x} dx + c \right) = e^{-2x} \left(xe^{2x} - \frac{1}{2}e^{2x} + c \right) = \left(x - \frac{1}{2} \right) + ce^{-2x}$
 $f(0) = 0 \Rightarrow f(0) = -\frac{1}{2} + c = 0, \quad c = \frac{1}{2} \Rightarrow f(x) = x - \frac{1}{2} + \frac{1}{2}e^{-2x}.$

51. 设 $f(x)$ 有一阶连续的导数, 并且满足

$$2 \int_0^x (x+1-t)f'(t)dt = x^2 - 1 + f(x),$$

求 $f(x)$.

解: $2 \int_0^x (x+1-t)f'(t)dt = x^2 - 1 + f(x) \Rightarrow 0 = \int_0^0 (0+1-t)f'(t)dt =$
 $0 - 1 + f(0) \Rightarrow f(0) = 1$.
 $2f'(x) + 2 \int_0^x f'(t)dt = 2x + f'(x) \Rightarrow f'(x) + 2f(x) - 2f(0) = 2x \Rightarrow$
 $f'(x) + 2f(x) = 2x + 2, \quad p(x) = 2, \quad e^{-\int p(x)dx} = e^{-2x}$.
 $f(x) = e^{-2x} \left(\int (2x+2)e^{2x} dx + c \right) = e^{-2x} \left(xe^{2x} - \frac{1}{2}e^{2x} + e^{2x} + c \right) = x + \frac{1}{2} + ce^{-2x}$.
 $f(0) = \frac{1}{2} + c = 1 \Rightarrow c = \frac{1}{2} \Rightarrow f(x) = x + \frac{1}{2} + \frac{1}{2}e^{-2x}.$

52. 设 $\varphi(x)$ 有一阶连续的导数, $\varphi(0) = 1$, 并设 $(y^2 + xy)dx + (\varphi(x) + 2xy)dy = 0$ 是全微分方程. 求 $\varphi(x)$ 及此全微分方程的通积分.

解: $(y^2 + xy)dx + (\varphi(x) + 2xy)dy = 0$ 是全微分方程.
 $\Rightarrow \frac{\partial(y^2 + xy)}{\partial y} = 2y + x, \quad \frac{\partial(\varphi(x) + 2xy)}{\partial x} = \varphi'(x) + 2y \Rightarrow \varphi'(x) = x,$

$$d(\varphi(x)) = xdx \Rightarrow \varphi(x) = \frac{1}{2}x^2 + c.$$

又 $\varphi(0) = c = 1 \Rightarrow \varphi(x) = \frac{1}{2}x^2 + 1 \Rightarrow (y^2 + xy)dx + \left(\frac{1}{2}x^2 + 2xy + 1\right)dy = 0$

$$\Rightarrow u(x, y) = \int_{x_0}^x (y_0^2 + xy_0)dx + \int_{y_0}^y \left(\frac{1}{2}x^2 + 2xy + 1\right)dy \\ = y_0^2x - y_0^2x_0 + \frac{1}{2}x^2y_0 - \frac{1}{2}x_0^2y_0 + \frac{1}{2}x^2y - \frac{1}{2}x_0^2y_0 + xy^2 - xy_0^2 + y + y_0 \\ \Rightarrow \frac{1}{2}x^2y + xy^2 + y = c.$$

用适当变换解下列方程 (53-55):

$$53. (x+y)^2 \frac{dy}{dx} = a^2.$$

解: 令 $z = x + y \Rightarrow z^2 \frac{dz - dx}{dx} = a^2 \Rightarrow z^2 \frac{dz}{dx} = a^2 + z^2 \Rightarrow \frac{z^2}{a^2 + z^2} dz = dx \Rightarrow (1 - \frac{a^2}{a^2 + z^2}) dz = dx \Rightarrow z - a \arctan \frac{z}{a} = x + c \Rightarrow x + y - a \arctan \frac{x + y}{a} = x + c \Rightarrow y = a \arctan \frac{x + y}{a} + c$.

$$54. \frac{dy}{dx} = y^2 - x^2 + 1 .$$

解: 令 $z = y - x \Rightarrow \frac{dz + dx}{dx} = z(z + 2x) + 1 \Rightarrow \frac{dz}{dx} = z^2 + 2xz \Rightarrow \frac{1}{z^2} \frac{dz}{dx} = 1 + \frac{2x}{z}$, 令 $u = \frac{1}{z} \Rightarrow -\frac{du}{dx} = 1 + 2xu \Rightarrow \frac{du}{dx} + 2xu = -1 \Rightarrow p(x) = 2x$,
 $e^{-\int p(x)dx} = e^{-x^2}$
 $\Rightarrow u = e^{-x^2} (\int -1 e^{x^2} dx + c)$
 $= e^{-x^2} (-\int e^{x^2} dx + c) = \frac{1}{z}$
 $\Rightarrow z = e^{x^2} (-\int e^{x^2} dx + c)^{-1} \Rightarrow y - x = e^{x^2} (-\int e^{x^2} dx + c)^{-1} \Rightarrow y = x + e^{x^2} (c - \int e^{x^2} dx)^{-1}$.

$$55. \frac{dy}{dx} = \frac{y}{2x} + \frac{1}{2y} \tan \frac{y^2}{x} .$$

解: $y \frac{dy}{dx} = \frac{y^2}{2x} + \frac{1}{2} \tan \frac{y^2}{x} \Rightarrow \frac{1}{2} \frac{dy^2}{dx} = \frac{y^2}{2x} + \frac{1}{2} \tan \frac{y^2}{x}$, 令 $z = \frac{y^2}{x} \Rightarrow \frac{d(xz)}{dx} = z + \tan z \Rightarrow \frac{x dz}{dx} + z = z + \tan z \Rightarrow x \frac{dz}{dx} = \tan z \Rightarrow \frac{\cos z}{\sin z} dz = \frac{1}{x} dx \Rightarrow \ln |\sin z| = \ln |x| + c \Rightarrow \sin z = cx \Rightarrow \sin \frac{y^2}{x} = cx \Rightarrow y^2 = x \arcsin cx$.

56. 求 $y = y'^2$ 的奇解。

解: $y' = p^2 \Rightarrow F(x, y, p) = p^2 - y = 0$, $\frac{\partial F}{\partial p} = 2p \Rightarrow p = 0 \Rightarrow y = 0$.

代入 $y = 0$ 是解 $\Rightarrow$ 是奇解。

57. 求 $y^2 y'^2 - 2xyy' + 2y^2 - x^2 = 0$ 的奇解。

解: $y' = p \Rightarrow F(x, y, p) = y^2 p^2 - 2xy p + 2y^2 - x^2 = 0$, $\frac{\partial F}{\partial p} = 2y^2 p - 2xy =$

$0 \Rightarrow (yp - x)y = 0$. $y = 0$ 代入显然不是上述方程的解。

$p = \frac{x}{y}$ 代入 $F(x, y, p) = 0 \Rightarrow x^2 - 2x^2 + 2y^2 - x^2 = 0$, $y = \pm x$.

$p = \pm 1$ 代入是方程的解 $\Rightarrow y = \pm x$ 是奇解。

58. 求 $[(y')^2 + 1](x - y)^2 = (x + yy')^2$ 的奇解。

解: $F = (p^2 + 1)(x - y)^2 - (x + yp)^2 = 0$, $\frac{\partial F}{\partial p} = 2p(x - y)^2 - 2(x + yp)y =$

$0 \Rightarrow p = \frac{y}{x - 2y} \Rightarrow y(x - y)^2(x - 2y) = 0 \Rightarrow$ 经检验 $y = 0$ 为奇解。

59. 求曲线族 $y = cx - (c^2 + 1)x^2$ 的包络, 其中 c 是参数。

解: $\frac{\partial \Phi}{\partial c} = x - 2cx^2 \Rightarrow c = \frac{1}{2x} \Rightarrow y = \frac{1}{4} - x^2 (x \neq 0)$ 。

60. 求曲线族 $\frac{x}{\sin \theta} + \frac{y}{\cos \theta} = a$ 的包络, 其中 a 是常数, θ 是参数。

解: $\frac{\partial \Phi}{\partial \theta} = -\frac{x \cos \theta}{\sin^2 \theta} + \frac{y \sin \theta}{\cos^2 \theta} = 0 \Rightarrow \frac{x}{\sin^3 \theta} = \frac{y}{\cos^3 \theta} \Rightarrow \frac{y \sin^2 \theta}{\cos^3 \theta} + \frac{y}{\cos \theta} =$
 $a \Rightarrow \frac{y}{\cos^3 \theta} = a \Rightarrow \frac{1}{\cos \theta} = \left(\frac{a}{y}\right)^{\frac{1}{3}}, \quad \frac{1}{\sin \theta} = \left(\frac{a}{x}\right)^{\frac{1}{3}} \Rightarrow x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}。$

61. 曲线族 $(y - a)^2 - x^3 = 0$ 有无包络? 其中 a 是参数。

解: $\frac{\partial \Phi}{\partial a} = 2(a - y) = 0 \Rightarrow y = a \Rightarrow c$ - 判别曲线 $x = 0$ 不是包络。

62. 求圆族 $(x - c)^2 + y^2 - \frac{b^2}{a^2}(a^2 - c^2) = 0$ 的包络, 其中 a, b 是常数, c 是参数。

解: $\frac{\partial \Phi}{\partial c} = 2(c - x) + 2c \frac{b^2}{a^2} = 0 \Rightarrow c = \frac{xa^2}{a^2 + b^2} \Rightarrow \frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1。$

求下列方程的解 (63-73):

63. $y' = \ln x。$

解: $y'' = \ln x \Rightarrow y' = x \ln x - x + c_1 \Rightarrow y'' = \frac{x^2}{2} \ln x - \frac{3}{4}x^2 + c_1x + c_2。$

64. $xy'' + y' = 4x。$

解: $y' = p \Rightarrow p'x + p = 4x \Rightarrow p = 2x + \frac{c}{x} \Rightarrow y = x^2 + c_1 \ln |x| + c_2。$

65. $2yy'' = (y')^2。$

解: $y' = p \Rightarrow 2yp \frac{dp}{dy} = p^2 \Rightarrow p = 0$ 或 $p = cy^{\frac{1}{2}} \Rightarrow y = c$ 或 $y = (c_1x + c_2)^2。$

66. $yy'' - (y')^2 = y^4, \quad y(0) = 1, \quad y'(0) = 0。$

解: $y' = p \Rightarrow yp \frac{dp}{dy} - p^2 = y^4 \Rightarrow \frac{dp^2}{dy} - \frac{2}{y}p^2 = 2y^3 \Rightarrow p^2 = y^2(c_1 + y^2) \Rightarrow$

$c_1 = -1 \Rightarrow \frac{d(\frac{1}{y})}{\sqrt{1 - (\frac{1}{y})^2}} = -dx \Rightarrow \frac{1}{y} = \sin(-x + c_2) \Rightarrow c_2 = \frac{\pi}{2} \Rightarrow y =$

$\sec x。$

67. $y'' = e^y。$

解: 令 $z = e^{\frac{1}{2}y} \Rightarrow y = 2 \ln z \Rightarrow \frac{dy}{dx} = \frac{2}{z} \frac{dz}{dx}, \quad \frac{dy^2}{dx^2} = \frac{2}{z} \frac{dz^2}{dx^2} - \frac{2}{z^2} \left(\frac{dz}{dx}\right)^2 \Rightarrow$
 $\frac{2}{z} z'' - \frac{2}{z} (z')^2 = z^2。$

$$\begin{aligned} \text{令 } p = z' \implies \frac{dp^2}{dz} - \frac{2}{z}p^2 &= z^3 \implies p^2 = z^2\left(\frac{1}{2}z^2 + c_1\right) \implies \frac{2c_1 dz}{z\sqrt{z^2 + c_1^2}} = \\ \sqrt{2}c_1 dx \implies \sqrt{2}c_1(x + c_2) &= \ln \frac{\sqrt{c_1^2 + e^y} - c_1}{\sqrt{c_1^2 + e^y} + c_1} . \end{aligned}$$

$$68. yy'' + (y')^2 = y' .$$

$$\text{解: } y' = p \implies yp \frac{dp}{dy} + p^2 = p \implies 1 - p = cy \implies y = c_1 + c_2 e^{-\frac{p}{c_1}} .$$

$$69. y^3 y'' + 1 = 0 .$$

$$\text{解: } p = y' \implies y^3 p \frac{dp}{dy} + 1 = 0 \implies \frac{p^2}{2} = \frac{1}{2y^2} + c \implies 1 + c_1 y^2 = (c_1 x + c_2)^2 .$$

$$70. 2y'' = 3y^2, \quad y(-2) = 1, \quad y'(-2) = 1 .$$

$$\begin{aligned} \text{解: } p = y' \implies 2p \frac{dp}{dy} &= 3y^2 \implies p^2 = y^3 + c \implies c = 0 \implies y = \frac{4}{(x + c_1)^2} \implies \\ c_1 = 0 \implies y &= \frac{4}{x^2} . \end{aligned}$$

$$71. y''(1 - y) + 2(y')^2 = 0 .$$

$$\text{解: } y' = p, \quad p \frac{dp}{dy} (1 - y) + 2p^2 = 0 \implies p = c(1 - y)^2 \implies \frac{1}{1 - y} = c_1 x + c_2 .$$

$$72. y'' + \sqrt{1 + (y')^2} = 0 .$$

$$\begin{aligned} \text{解: } y' = p, \quad p' + \sqrt{1 + p^2} &= 0 \implies y' + \sqrt{1 + (y')^2} = ce^{-x} \implies y' = \\ \frac{1}{2}ce^{-x} - \frac{1}{2c}e^{-x} \implies y &= \frac{1}{2}c_1 e^{-x} - \frac{1}{2c_1}e^x + c_2 . \end{aligned}$$

$$73. xy'' = y' \ln \frac{y'}{x} .$$

$$\begin{aligned} \text{解: 令 } y' = p \implies xp' &= p \ln \frac{p}{x} . \text{ 令 } z = \ln \frac{p}{x} \implies p = xe^z, \quad \frac{dp}{dx} = e^z + xe^z \frac{dz}{dx} \implies \\ x(e^z + xe^z \frac{dz}{dx}) &= zxe^z \implies x \frac{dz}{dx} - z = -1 \implies z = x(c_1 + \frac{1}{x}) = c_1 x + 1 \implies y' = \\ exe^{c_1 x} . \end{aligned}$$

$$\text{当 } c_1 = 0 \text{ 时, } y = \frac{1}{2}ex^2 + c ,$$

$$\text{当 } c_1 \neq 0 \text{ 时, } y = \frac{e}{c_1}(x - \frac{1}{c_1})e^{c_1 x} + c_2 .$$

74. 设当 $x \geq 0$ 时 $f(x)$ 有一阶连续导数, 并且满足

$$f(x) = -1 + x + 2 \int_0^x (x - t)f(t)f'(t)dt ,$$

求 $f(x)$ (当 $x \geq 0$) .

$$\begin{aligned} \text{解: } f'(x) &= 1 + 2 \int_0^x f(t)f'(t)dt \\ \implies f'(x) &= 1 + f^2(x) - f^2(0) = f^2 \quad (f(0) = -1) \\ \implies -\frac{1}{f} &= x + c, \quad c = 1 \end{aligned}$$

$$\Rightarrow f(x) = -\frac{1}{x+1}.$$

75. 设曲线通过点 $A(1, -1)$ ，且曲线上任一点处的切线斜率等于切点纵坐标的平方，求此曲线的方程。

$$\text{解: } y(1) = -1, \quad y' = y^2 \Rightarrow -\frac{1}{y} = x + c \Rightarrow c = 0 \Rightarrow y = -\frac{1}{x}.$$

76. 设 100 摄氏度的物体置于 20 摄氏度的屋子里，在 10 分钟内冷却到 60 摄氏度，问在多少时间内该物体冷却到 25 摄氏度。

$$\text{解: } y' = k(y - 20) \Rightarrow \ln y - 20 = kt + c, \quad y(0) = 100, \quad y(10) = 60 \Rightarrow c = \ln 80, \quad k = -\frac{1}{10} \ln 2 \Rightarrow \text{当 } y(t) = 25 \text{ 时, } t = 40m.$$

77. 已知放射性物质镭的裂变规律是：裂变速率与剩余量成正比。设已知在某一时刻 $t = t_0$ 时，镭的份量是 R_0 克，求在任意时刻 t 镭的份量 $R(t)$ 。

$$\text{解: } R'(t) = -\lambda R(t), \quad R(t_0) = R_0 \Rightarrow R(t) = ce^{-\lambda(t-t_0)}, \quad c = R_0 \Rightarrow R(t) = R_0 e^{-\lambda(t-t_0)}.$$

78. 一厂房体积为 V 立方米，开始时空气中含有二氧化碳 m_0 克，每分钟通入体积为 Q 立方米的新鲜空气（设新鲜空气中不含二氧化碳），同时排出等量的混浊空气，室内空气始终保持均匀，求室内二氧化碳的含量与时间的函数关系。

$$\text{解: } y(0) = m_0, \quad y' = -\frac{Q}{V}y \Rightarrow y = ce^{-\frac{Q}{V}t} \Rightarrow c = m_0 \Rightarrow y = m_0 e^{-\frac{Q}{V}t}.$$

79. 已知曲线的曲率处处都等于常数 $k(k \neq 0)$ ，法线方程为 $-y'(Y - y) = X - x$ ， $Y = 0$ 时， $X = x + yy'$ ，求此曲线的方程。

$$\text{解: 曲线过 } Q(x + yy', 0) \text{ 点, } |PQ| = \sqrt{(x + yy' - x)^2 + y^2} = \sqrt{y^2 p^2 + y^2} = \frac{1}{k}, \quad y^2 p^2 + y^2 = \frac{1}{k^2} \Rightarrow p^2 = \frac{1}{y^2 k^2} - 1 \Rightarrow p = \pm \sqrt{\frac{1 - y^2 k^2}{y^2 k^2}} \Rightarrow \frac{dy}{dx} = \pm \sqrt{\frac{1 - y^2 k^2}{y^2 k^2}}.$$

$$\text{取 } + \text{ 时, 不妨先设 } y > 0 \Rightarrow \frac{yk}{\sqrt{1 - y^2 k^2}} dy = dx \Rightarrow -\frac{1}{2k} \frac{d(1 - y^2 k^2)}{\sqrt{1 - y^2 k^2}} = dx \Rightarrow -\frac{1}{k} \sqrt{1 - y^2 k^2} = x + c \Rightarrow 1 - y^2 k^2 = k^2(x + c)^2 \Rightarrow \frac{1}{k^2} = (x + c)^2 + y^2 \Rightarrow \text{是圆, 其余几种情况类似可得都是圆.}$$

80. 求一曲线族，使在其上每一点处与曲线族 $y = cx^3$ 正交。

$$\text{解: } y' = \frac{-1}{3cx^2} = -\frac{x}{3y} \Rightarrow x^2 + 3y^2 = c.$$

81. 一盛满水的直立圆柱形贮水器，直径为 4 米，高为 6 米，其底上有

一半径为 $\frac{1}{12}$ 米的圆孔, 问容器中水全部由小孔流完需多少时间? 已知水从小孔流出的速度等于 $0.6\sqrt{2gh}$ (g 是重力加速度, h 是小孔离液面的距离)。

$$\text{解: } h' = -\frac{(\frac{1}{12})^2 0.6\sqrt{2gh}}{2^2} \Rightarrow 2\sqrt{h} = \frac{-0.6t\sqrt{2g}}{24^2} + c,$$

$$h(0) = 6 \Rightarrow c = 2\sqrt{6}.$$

$$\text{当 } h = 0 \text{ 时, } t = 1062s = 17.7m.$$

82. 设对任意 $x > 0$, 曲线 $y = f(x)$ 上点 $(x, f(x))$ 处的切线在 y 轴上的截距等于 $\frac{1}{x} \int_0^x f(t)dt$, 求 $f(x)$ 的一般表达式。

$$\text{解: } y - f(x_0) = f'(x_0)(x - x_0) \Rightarrow \frac{1}{x_0} \int_0^{x_0} f(t)dt - f(x_0) = -f'(x_0)x_0 \Rightarrow xf''(x) = -f'(x) \Rightarrow f(x) = c_1 \ln x + c_2.$$

83. 某湖泊的水量为 V , 每年排入湖泊内含污染物 A 的污水量为 $\frac{V}{6}$, 流入湖泊内不含 A 的水量为 $\frac{V}{6}$, 流出湖泊的水量为 $\frac{V}{3}$ 。已知 1999 年底湖中 A 的含量为 $5m_0$, 超过了国家规定指标。为了治理污染, 从 2000 年初起, 限定排入湖泊中含 A 污水的浓度不得超过 $\frac{m_0}{V}$ 。问至多经过多少年, 湖泊中污染物 A 的含量就可降至 m_0 以内? (注: 设湖水中 A 的浓度是均匀的。)

$$\text{解: } m' = \frac{m_0}{6} - \frac{m}{3} \Rightarrow m(t) = \frac{m_0}{2} + ce^{-\frac{t}{3}},$$

$$\text{又 } m(0) = 5m_0 \Rightarrow c = \frac{9}{2}m_0 \Rightarrow \text{当 } m(t) = m_0 \text{ 时, } t = 6 \ln 3.$$

84. 求一条凹曲线, 已知其上任一点处的曲率 $k = \frac{1}{2y^2 \cos \alpha}$, 其中 α 为该曲线在相应点处的切线的倾角 ($\cos \alpha > 0$), 且曲线在点 $(1, 1)$ 处的切线为水平。

$$\text{解: } \frac{y''}{(1 + (y')^2)^{\frac{3}{2}}} = \frac{\sqrt{1 + (y')^2}}{2y^2}.$$

$$\text{令 } y' = p, \quad y(1) = 1, \quad y'(1) = 0, \quad \frac{1}{y} = \frac{1}{1 + p^2} + c \Rightarrow c = 0 \Rightarrow p = \sqrt{y-1} \Rightarrow 4y = (x + c_1)^2 + 1 \Rightarrow c_1 = -1 \Rightarrow 4y = (x-1)^2 + 4.$$

85. 求连接两点 $A(0, 1)$ 与 $B(1, 0)$ 的一条曲线, 它位于弦 AB 的上方, 并且对于此弧上的任意一条弦 AP, 该曲线与弦 AP 之间的面积为 x^3 , 其中 x 为点 P 的横坐标。

$$\text{解: } \int_0^x y(t)dt - \left(\frac{y(x)-1}{x}t + 1\right)dt = x^3, \quad x \in [0, 1]$$

$$\Rightarrow \int_0^x y(t)dt - \frac{x}{2}(y-1) - x = x^3 \Rightarrow y - xy' = 6x^2 + 1 \Rightarrow y'' = -12 \Rightarrow y = -6x^2 + c_1x + c_2, \quad y(0) = 1, \quad y(1) = 0 \Rightarrow c_1 = 5, \quad c_2 = 1 \Rightarrow y = -6x^2 + 5x + 1.$$

86. 跳伞运动员从高空自飞机上跳下, 经若干秒后打开降落伞, 开伞后运动过程中所受空气阻力为 kv^2 , 其中常数 $k > 0$, v 为下降速度, 设人与伞的质量为 m , 且不计空气浮力, 试证明: 只要打开伞后有足够的降落时间着地, 则落地速度将近似地等于 $\sqrt{\frac{mg}{k}}$ 。

解: $mv' = mg - kv^2$ (当时间足够时, $v' = 0$, 即 $v = \sqrt{\frac{mg}{k}}$)。
 $v' = -\frac{k}{m}(v^2 - \frac{gm}{k})$, 令 $b^2 = \frac{gm}{k}$, $a^2 = \frac{kg}{m} \Rightarrow \frac{v-b}{v+b} = ce^{-\frac{2k}{m}bt} = ce^{-2at} \Rightarrow$
 $v = b \frac{ce^{2at+1}}{ce^{2at-1}} \Rightarrow$ 当 t 充分大时, $v = b = \sqrt{\frac{mg}{k}}$ 。

87. 设函数 $p(x)$ 和 $f(x)$ 在区间 $[0, +\infty)$ 上连续, 且 $\lim_{x \rightarrow +\infty} p(x) = a > 0$, $|f(x)| \leq b$, a, b 均为常数。试证明: 方程 $\frac{dy}{dx} + p(x)y = f(x)$ 的一切解在 $[0, +\infty)$ 上有界。

解: $p(x), f(x)$ 在 R^+ 上连续 $\Rightarrow y = e^{-\int p(t)dt} [c + \int f(t)e^{\int p(s)ds} dt]$ 在 R^+ 也连续。

又 $\lim_{x \rightarrow +\infty} p(x) = a > 0, |f(x)| \leq b \Rightarrow \lim_{x \rightarrow +\infty} ce^{-\int_0^x p(t)dt} = 0$, 即 $ce^{-\int_0^x p(t)dt}$ 在 R^+ 上有界。

而 $\lim_{x \rightarrow +\infty} |e^{-\int p(t)dt} \int f(t)e^{\int p(s)ds}| \leq b \lim_{x \rightarrow +\infty} |\frac{\int e^{\int p(s)ds} dt}{e^{\int p(t)dt}}| = b \lim_{x \rightarrow +\infty} |\frac{e^{\int p(s)ds} dt}{p(x)e^{\int p(t)dt}}| = \frac{b}{a}$
 $\Rightarrow e^{-\int p(t)dt} \int f(t)e^{\int p(s)ds} dt$ 在 R^+ 上也有界 $\Rightarrow y$ 在 R^+ 上有界。

88. 设初值问题

$$\begin{cases} x \frac{dy}{dx} - (2x^2 + 1)y = x^2, & x \geq 1, \\ y(1) = y_1. \end{cases}$$

(1) 求满足上述初值问题的解 (用积分表示);

(2) 是否存在适当的 y_1 , 使对应的解 $y(x)$ 当 $x \rightarrow +\infty$ 时存在有限极限? 若有, 这种 y_1 有多少? 求出之, 并求 $\lim_{x \rightarrow +\infty} y(x)$ 。

解: $y' - (2x + \frac{1}{x})y = x$
 $\Rightarrow y = e^{x^2 + \ln x} [c + \int_1^x te^{-t^2 - \ln t} dt] = xe^{x^2} [c + \int_1^x e^{-t^2} dt]$ 。
 由 $y(1) = y_1 \Rightarrow c = e^{-1}y_1 \Rightarrow y = xe^{x^2} [y_1 e^{-1} + \int_1^x e^{-t^2} dt]$ 。

当 $\lim_{x \rightarrow +\infty} y$ 存在时,

$\lim_{x \rightarrow +\infty} y' = 0 \Rightarrow \lim_{x \rightarrow +\infty} (2 + \frac{1}{x^2})y = -1$, 即 $\lim_{x \rightarrow +\infty} y = -\frac{1}{2}$ 。

$$\text{又 } \lim_{x \rightarrow +\infty} \frac{(y_1 e^{-1} + \int_1^x e^{-t^2} dt)'}{(\frac{1}{x} e^{-x^2})'} = \lim_{x \rightarrow +\infty} \frac{e^{-x^2}}{-\frac{1}{x^2} e^{-x^2} - \frac{1}{x} 2x e^{-x^2}} = -\frac{1}{2}$$

$$\Rightarrow \text{要使 } y \text{ 极限存在只需 } \lim_{x \rightarrow +\infty} y_1 e^{-1} + \int_1^x e^{-t^2} dt = 0, \text{ 即 } y_1 = -e \int_1^{+\infty} e^{-t^2} dt.$$

89. 求 $y' + y \cos x = \sin x$ 的通解 (用积分表示); 在这些解中, 有无周期为 2π 的? 若有, 求出之, 若无, 说明理由。

$$\text{解: } y = e^{\sin x} [c + \int_0^x \sin t e^{-\sin t} dt],$$

此解不以 2π 为周期 (因为 $\int_0^{2\pi} \sin t e^{-\sin t} dt \neq 0$)。

第二章 线性微分方程 习题解答

2004 年 10 月 10 日

1. 证明: 设函数 $f_1(x), f_2(x), \dots, f_m(x)$ 在区间 (a, b) 内线性无关, 则这些函数中的部分函数在区间 (a, b) 内也线性无关. 换句话说, 如果函数 $f_1(x), f_2(x), \dots, f_m(x)$ 在区间 (a, b) 内线性相关, 则添上一些函数也是线性相关.

证明: 若函数 $f_1(x), f_2(x), \dots, f_m(x)$ 线性无关, 则若函数 $f_1(x), f_2(x), \dots, f_k(x)$ 线性相关, 那么存在 $c_1, c_2, \dots, c_k$, 使得函数 $c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x) \equiv 0$, 其中 $c_1, c_2, \dots, c_k$ 不全为 0。不妨设 $c_1 \neq 0$, 那么函数 $c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x) + c_{k+1} f_{k+1}(x) + \dots + c_m f_m(x) \equiv 0$, 其中 $c_{k+1} = c_{k+2} = \dots = c_m = 0$ 。但因为 $c_1 \neq 0$, 所以函数 $f_1(x), f_2(x), \dots, f_m(x)$ 线性相关, 产生矛盾! 所以就可以得到函数 $f_1(x), f_2(x), \dots, f_m(x)$ 线性无关。

2. 证明: 设函数 $f_1(x), f_2(x), \dots, f_k(x)$ 在 (a, b) 内线性无关, 则由这些函数构造出的 k 个新的函数

$$g_i(x) = \sum_{j=1}^k a_{ij} f_j(x) \quad (i = 1, 2, \dots, k),$$

在 (a, b) 内也线性无关的充分必要条件是行列式

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{vmatrix} \neq 0$$

证明: " $\Rightarrow$ " $g_1(x), g_2(x), \dots, g_k(x)$ 线性无关, 则对于任意的 $c_1, c_2, \dots, c_k$, 我们有 $c_1 g_1 + c_2 g_2 + \dots + c_k g_k \equiv 0 \Leftrightarrow c_1 = c_2 = \dots = c_k = 0$ 。而 $g_i(x) = \sum_{j=1}^k a_{ij} f_j(x)$, 所以 $\sum_{i=1}^k c_i \sum_{j=1}^k a_{ij} f_j = \sum_{j=1}^k \sum_{i=1}^k c_i a_{ij} f_j = \sum_{j=1}^k (\sum_{i=1}^k c_i a_{ij}) f_j \equiv 0$ 。因为 $f_1(x), f_2(x), \dots, f_k(x)$ 线性无关, 所以 $\sum_{i=1}^k c_i a_{ij} = 0$ 。即

[illegible]

而由上可知其只有零解。所有由代数知识, 我们有

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{vmatrix} \neq 0$$

” $\Leftarrow$ ”和前面一样,若 $c_1g_1 + c_2g_2 + \cdots + c_kg_k \equiv 0$, 我们有 $\sum_{i=1}^k c_i \sum_{j=1}^k a_{ij}f_j = 0$, 即

[illegible]

同样由代数知识可知由于

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{vmatrix} \neq 0$$

所以方程只有零解, 也就是 $c_1 = c_2 = \cdots = c_k = 0$, 即得 $g_1, g_2, \cdots, g_k$ 线性无关。

3. 证明: 设函数 $f_1(x), f_2(x), \dots, f_k(x), f_{k+1}(x), \dots, f_{k+m}(x)$ 在 (a, b) 内线性无关, 并假设由其中的部分函数, 例如 $f_1(x), f_2(x), \dots, f_k(x)$ 构造 k 个新的函数

$$g_i(x) = \sum_{j=1}^k a_{ij} f_j(x) \quad (i = 1, 2, \dots, k)$$

在 (a, b) 内线性无关, 则以 $g_1(x), g_2(x), \dots, g_k(x)$ 代替 $f_1(x), f_2(x), \dots, f_k(x)$ 得到的 $k+m$ 个函数 $g_1(x), g_2(x), \dots, g_k(x), f_{k+1}(x), \dots, f_{k+m}(x)$ 在区间 (a, b) 内也线性无关。

证明: " $\Leftarrow$ " 定义 $g_{k+1} = f_{k+1}, \cdots, g_{k+m} = f_{k+m}$, 也即 $a_{ii} = 1, i \geq k+1, a_{ij} = 0, i \geq k+1$, 所以

$$A = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & a_{2k} & 0 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 1 \end{vmatrix}$$

$$= (-1)^{2(k+m+\cdots+k+1)} \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{vmatrix}$$

所以 $f_1, \dots, f_{k+m}$ 线性无关, 所以 $f_1, \dots, f_k$ 也线性无关。则由 1 得 $c_1 f_1 + \dots + c_k f_k = 0$, 有某个 $c_i \neq 0$ 。那么 $c_1 f_1 + \dots + c_k f_k + 0 \cdot f_{k+1} + \dots + 0 \cdot f_{k+m} = 0 \Rightarrow f_1, \dots, f_{k+m}$ 也线性相关。产生矛盾。所以, $f_1, \dots, f_k$ 线性无关。由 2 得

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{vmatrix} \Rightarrow A \neq 0$$

由 2 得 $g_1, \cdots, g_{k+m}$ 线性无关。由因为 $f_{k+i} = g_{k+i}$ ，所以 $g_1, \cdots, g_k, f_{k+1}, \cdots, f_{k+m}$ 线性无关

4. 设 $y_i (i = 1, \cdots, n+1)$ 是 n 阶非齐次线性方程 $L[y] = f(x)$ 的 $n+1$ 个线性无关的解, 试求对应的齐次线性方程 $L[y] = 0$ 的基本解组; 并求 $L[y] = f(x)$ 的通解。

解: 令 $z_i = y_{i+1} - y_i$, 那么 $L(z_i) = L(y_{i+1}) - L(y_i) = f(x) - f(x) = 0$ 。所以 $z_i, i = 1, 2, \dots, n$ 是齐次线性方程 $L[y] = 0$ 的一组解。又设 $c_1 z_1 + c_2 z_2 + \dots + c_n z_n = 0$, 那么代入

后, $c_n y_{n+1} + (c_{n-1} - c_n) y_n + \cdots + (c_1 - c_2) y_2 - c_1 y_1 = 0$ 。因为 $y_1, y_2, \cdots, y_{n+1}$ 线性无关, 所以 $c_n = 0, c_{n-1} - c_n = 0, \cdots, c_{i-1} - c_i = 0, \cdots, c_1 - c_2 = 0, c_1 = 0 \Rightarrow c_{i-1} = c_i \Rightarrow c_1 = c_2 = \cdots = c_n = 0$, 那么我们就有 $z_1, z_2, \cdots, z_n$ 线性无关。所以 $z_1, z_2, \cdots, z_n$ 是 $L[y] = 0$ 的基本解组。那么 $L[y] = f(x)$ 的通解为

$$y = c_1 z_1 + c_2 z_2 + \cdots + c_n z_n + y_1$$

$$= (1 - c_1) y_1 + (c_1 - c_2) y_2 + \cdots + (c_{n-1} - c_n) y_n + c_n y_{n+1}, \forall c_1, \cdots, c_n$$

5. 设 $y_i (i = 1, \cdots, n+1)$ 是齐次线性方程

$$y^n + p_1(x) y^{n-1} + p_2(x) y^{n-2} + \cdots + p_n(x) y = 0$$

的基本解组, 其中 $p_i(x) (i = 1, \cdots, n+1)$ 在区间 (a, b) 内连续。 $W(x)$ 是 $y_1(x), y_2(x), \cdots, y_n(x)$ 的朗斯基行列式。试证明下述刘维尔公式成立:

$$W(x) = W(x_0) \exp\left[-\int_{x_0}^x p_1(\xi) d\xi\right], \quad x_0, x \in (a, b).$$

其中 $\exp u = e^u$ 。

证明:

$$W = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \cdots & \cdots & \cdots & \cdots \\ y_1^{n-1} & y_2^{n-1} & \cdots & y_n^{n-1} \end{vmatrix}$$

$$W' = \begin{vmatrix} y'_1 & y'_2 & \cdots & y'_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \cdots & \cdots & \cdots & \cdots \\ y_1^{n-1} & y_2^{n-1} & \cdots & y_n^{n-1} \end{vmatrix} + \cdots + \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \cdots & \cdots & \cdots & \cdots \\ y_1^n & y_2^n & \cdots & y_n^n \end{vmatrix}$$

$$= \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \cdots & \cdots & \cdots & \cdots \\ y_1^n & y_2^n & \cdots & y_n^n \end{vmatrix}$$

$$= \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \cdots & \cdots & \cdots & \cdots \\ -\sum_{i=1}^n i y_1^{n-i} & -\sum_{i=1}^n i y_2^{n-i} & \cdots & -\sum_{i=1}^n i y_n^{n-i} \end{vmatrix}$$

$$= -\sum_{i=1}^n p_i \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \cdots & \cdots & \cdots & \cdots \\ y_1^{n-i} & y_2^{n-i} & \cdots & y_n^{n-i} \end{vmatrix}$$

$$= -p_1 \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y'_1 & y'_2 & \cdots & y'_n \\ \cdots & \cdots & \cdots & \cdots \\ y_1^{n-1} & y_2^{n-1} & \cdots & y_n^{n-1} \end{vmatrix} = -p_1 W$$

所以 $W' = -p_1 W$, 即 $\frac{W'}{W} = -p_1$ 那么就有 $\int_{x_0}^x \frac{W'}{W} dx = -\int_{x_0}^x p_1$, 则 $\ln W(x) - \ln W(x_0) = -\int_{x_0}^x p_1(x) dx$, 于是就得到 $W(x) = W(x_0) \exp\left[-\int_{x_0}^x p_1(\xi) d\xi\right]$

求下列方程的通解或特解 (6~ 17):

$$6.4y' - 3y = 0.$$

解: $4\lambda - 3 = 0, \lambda = \frac{3}{4}$, 通解为 $y = ce^{\frac{3}{4}x}$.

$$7.y'' - 4y' = 0.$$

解: $\lambda^2 - 4\lambda = 0, \lambda_1 = 0, \lambda_2 = 4$, 通解为 $y = c_1 + c_2e^{4x}$.

$$8.y'' + 2y = 0.$$

解: $\lambda^2 + 2 = 0, \lambda_1 = \sqrt{2}i, \lambda_2 = -\sqrt{2}i$, 通解为 $y = c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x$.

$$9.y'' - 2y' + y = 0.$$

解: $\lambda^2 - 2\lambda + 1 = 0 \quad \lambda = 1$, 通解为 $y = (c_1 + c_2x)e^x$.

$$10.y'' + 4y' + 13y = 0.$$

解: $\lambda^2 + 4\lambda + 13 = 0, \lambda_1 = -2 + 3i, \lambda_2 = -2 - 3i$, 通解为 $y = e^{-2x}(c_1 \cos 3x + c_2 \sin 3x)$.

$$11.y'' - 5y' + 4y = 0, y|_{x=0} = 5, y'|_{x=0} = 8.$$

解: $\lambda^2 - 5\lambda + 4 = 0, \lambda_1 = 1, \lambda_2 = 4$, 则通解为 $y = c_1e^x + c_2e^{4x}$, 于是我们有 $y' = c_1e^x + 4c_2e^{4x}$,

代入初始条件 $\begin{cases} c_1 + c_2 = 5 \\ c_1 + 4c_2 = 8 \end{cases}$, 于是有 $\begin{cases} c_1 = 4 \\ c_2 = 1 \end{cases}$, 那么解为: $y = 4e^x + e^{4x}$.

$$12.y''' - 2y'' - y' + 2y = 0.$$

解: $\lambda^3 - 2\lambda^2 - \lambda + 2 = 0, \lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 3$, 通解为 $y = c_1e^{-x} + c_2e^x + c_3e^{2x}$.

$$13.y''' - y'' + y' - y = 0.$$

解: $\lambda^3 - \lambda^2 + \lambda - 1 = 0, \lambda_1 = 1, \lambda_2 = i, \lambda_3 = -i$, 通解为 $y = c_1e^x + c_2 \cos x + c_3 \sin x$.

$$14.y^{(4)} + 2y'' + y = 0.$$

解: $\lambda^4 + 2\lambda^2 + 1 = 0, \lambda_{1,2} = i, \lambda_{3,4} = -i$, 通解为 $y = (c_1 + c_2x) \cos x + (c_3 + c_4x) \sin x$.

$$15.y^{(4)} + 4y''' + 8y'' + 8y' + 4y = 0.$$

解: $\lambda^4 + 4\lambda^3 + 8\lambda^2 + 8\lambda + 4 = 0, \lambda_{1,2} = -1 + i, \lambda_{3,4} = -1 - i$, 通解为 $y = e^{-x}((c_1 + c_2x) \cos x + (c_3 + c_4x) \sin x)$.

$$16.y^{(4)} - 4y''' + 6y'' - 4y' + y = 0.$$

解: $\lambda^4 - 4\lambda^3 + 6\lambda^2 - 4\lambda + 1 = 0, \lambda_{1,2,3,4} = 1$, 通解为 $y = (c_1 + c_2x + c_3x^2 + c_4x^3)e^x$.

$$17.y^{(4)} - y = 0, y(0) = 2, y'(0) = -1, y''(0) = -2, y'''(0) = 1.$$

解: $\lambda^4 - 1 = 0, \lambda_1 = 1, \lambda_2 = -1, \lambda_3 = i, \lambda_4 = -i$, 通解为 $y = c_1e^x + c_2e^{-x} + c_3 \cos x + c_4 \sin x$, 于是有 $y' = c_1e^x - c_2e^{-x} - c_3 \sin x + c_4 \cos x$, $y'' = c_1e^x + c_2e^{-x} - c_3 \cos x - c_4 \sin x$, $y''' = c_1e^x -$

$c_2e^{-x} + c_3 \sin x - c_4 \cos x$, 代入初始条件 $\begin{cases} c_1 + c_2 + c_3 = 2 \\ c_1 - c_2 + c_4 = -1 \\ c_1 + c_2 - c_3 = -2 \\ c_1 - c_2 - c_4 = 1 \end{cases}$, 则有 $\begin{cases} c_1 = 0 \\ c_2 = 0 \\ c_3 = 2 \\ c_4 = -1 \end{cases}$, 于是解为

$$y = 2 \cos x - \sin x.$$

求下列方程的通解或特解 (18~36):

$$18.y'' + y = a(a \text{ 是常数}), y(0) = 0, y'(0) = 0.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 \cos x + c_2 \sin x$, 去特解 $y_0 = A$, 则 $A = a$, 所以 $y = c_1 \cos x + c_2 \sin x + a, y' = -c_1 \sin x + c_2 \cos x$, 代入初值 $\begin{cases} c_1 + a = 0 \\ c_2 = 0 \end{cases}$, 得到 $\begin{cases} c_1 = -a \\ c_2 = 0 \end{cases}$, 于是解为 $y = -a \cos x + a$

$$19.y'' + 5y' + 4y = 20e^x, y(0) = 0, y'(0) = -2.$$

解: 齐次方程的通解为 $\tilde{y} = c_1e^{-x} + c_2e^{-4x}$, 设特解为 $y_0 = Ae^x$, 则 $Ae^x + 5Ae^x + 4Ae^x = 20e^x$, 就可以得到 $A = 2$. 于是 $y = c_1e^{-x} + c_2e^{-4x} + 2e^x$, $y' = -c_1e^{-x} - 4c_2e^{-4x} + 2e^x$, 代入初始条件, 我们有 $\begin{cases} c_1 + c_2 + 2 = 0 \\ -c_1 - 4c_2 + 2 = -2 \end{cases}$, 得到 $\begin{cases} c_1 = -4 \\ c_2 = 2 \end{cases}$, 那么解就为 $y = -4e^{-x} + 2e^{-4x} + 2e^x$.

$$20.y'' + y = xe^{-x}.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 \cos x + c_2 \sin x$, 设特解为 $y_0 = (Ax + B)e^{-x}$, 则 $y_0'' = (Ax - 2A + B)e^{-x}$, 所以 $(2Ax - 2A + 2B)e^{-x} = xe^{-x}$, 那么 $A = \frac{1}{2}, B = \frac{1}{2}$, 所以就得到解为 $y =$

$$c_1 \cos x + c_2 \sin x + (\frac{1}{2}x + \frac{1}{2})e^{-x}.$$

$$21. y'' + 6y' + 5y = -10x + 8.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 e^{-x} + c_2 e^{-5x}$, 设特解为 $y_0 = Ax + B, y'_0 = A, y''_0 = 0$, 所以就有 $6A + 5(Ax + B) = -10x + 8$, 则 $A = -2, B = 4$, 于是解为 $y = c_1 e^{-x} + c_2 e^{-5x} - 2x + 4$.

$$22. y' + 4y = x^2.$$

解: 齐次方程的通解为 $\tilde{y} = ce^{-4x}$, 设特解为 $y_0 = Ax^2 + Bx + C$, 则有 $y'_0 = 2Ax + B$, 所以可以得到 $A = \frac{1}{4}, B = -\frac{1}{8}, C = \frac{1}{32}$, 那么解就为 $y = Ce^{-4x} + \frac{1}{4}x^2 - \frac{1}{8}x + \frac{1}{32}$.

$$23. y'' + 4y' + 1 = 0.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 + c_2 e^{-4x}$, 设特解为 $y_0 = Ax$, 于是有 $y'_0 = A$, 则可得 $A = -\frac{1}{4}$, 那么就可以得到解为 $y = c_1 + c_2 e^{-4x} - \frac{1}{4}x$.

$$24. y'' + 2y' + y = 2e^{-x}.$$

解: 齐次方程的通解为 $\tilde{y} = (c_1 + c_2 x)e^{-x}$, 设特解为 $y_0 = Ax^2 e^{-x}$, 于是有 $y'_0 = 2Axe^{-x} - Ax^2 e^{-x}$, $y''_0 = 2Ae^{-x} - 4Axe^{-x} + Ax^2 e^{-x}$, 则有 $A = 1$, 那么解为 $y = (c_1 + c_2 x + x^2)e^{-x}$.

$$25. y'' - 4y = e^{2x}, y|_{x=0} = 1, y'|_{x=0} = 2.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 e^{-2x} + c_2 e^{2x}$, 设特解为 $y_0 = Axe^{2x}$, 于是有 $y''_0 = (4A + 4Ax)e^{2x}$, 则有 $A = \frac{1}{4}$, 那么有 $y = c_1 e^{-2x} + c_2 e^{2x} + \frac{1}{4}xe^{2x}$, 则有 $y' = -2c_1 e^{-2x} + 2c_2 e^{2x} + \frac{1}{4}xe^{2x} + \frac{1}{2}e^{2x}$, 代入初值条件 $\begin{cases} c_1 + c_2 = 1 \\ -2c_1 + 2c_2 + \frac{1}{4} = 2 \end{cases}$, 得到 $\begin{cases} c_1 = \frac{1}{16} \\ c_2 = \frac{15}{16} \end{cases}$, 于是得到解为 $y = \frac{1}{16}e^{-2x} + (\frac{15}{16} + \frac{1}{4}x)e^{2x}$.

$$26. \frac{d^2 x}{dt^2} + x = \cos 2t, x|_{t=0} = \frac{dx}{dt}|_{t=0} = -2.$$

解: 齐次方程的通解 $\tilde{x} = c_1 \cos t + c_2 \sin t$, 设特解为 $x_0 = Ae^{2it}$, 则有 $x''_0 = -4Ae^{2it}$, 于是就得到 $A = -\frac{1}{3}$, 所以 $x = c_1 \cos t + c_2 \sin t - \frac{1}{3} \cos 2t$, $x' = -c_1 \sin t + c_2 \cos t + \frac{2}{3} \sin 2t$, 代入初始条件就得到 $\begin{cases} c_1 - \frac{1}{3} = -2 \\ c_2 = -2 \end{cases}$, 得到 $\begin{cases} c_1 = -\frac{5}{3} \\ c_2 = -2 \end{cases}$, 于是解为 $x = -\frac{5}{3} \cos t - 2 \sin t - \frac{1}{3} \cos 2t$.

$$27. \frac{d^2 x}{dt^2} + x = \sin at, a > 0.$$

解: 齐次方程的通解 $\tilde{x} = c_1 \cos t + c_2 \sin t$

$a \neq 1$ 时: 设特解为 $x_0 = Ae^{ait}$, 此时有 $x''_0 = -a^2 Ae^{ait}$, 可得 $A = \frac{1}{1-a^2}$, 于是有 $x = c_1 \cos t + c_2 \sin t + \frac{1}{1-a^2} \sin at$.

$a = 1$ 时: 设特解为 $x_0 = Ate^{it}$, 此时有 $x''_0 = (2Ai - At)e^{it}$, 可得 $A = -\frac{1}{2}$, 于是就有解为 $x = c_1 \cos t + c_2 \sin t - \frac{1}{2}t \sin t$.

$$28. \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} = 2 \sin x + \cos x.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 + c_2 e^{-3x}$, 设特解为 $y_0 = A \sin x + B \cos x$, 可得 $y'_0 = A \cos x - B \sin x, y_0 = -A \sin x - B \cos x$, 于是就可得到 $A = \frac{1}{10}, B = -\frac{7}{10}$, 那么解就为 $y = c_1 + c_2 e^{-3x} + \frac{1}{10} \sin x - \frac{7}{10} \cos x$.

$$29. \frac{d^2 x}{dt^2} + 2k \frac{dx}{dt} + 2k^2 x = 5k^2 \sin kt.$$

解: $k = 0$ 时: 为齐次方程, $x = c_1 t + c_2$;

$k \neq 0$ 时: 齐次方程的通解为 $\tilde{x} = e^{-kt}(c_1 \cos kx + c_2 \sin kx)$, 设特解为 $x_0 = Ae^{kit}$, 则有 $x'_0 = Akie^{kit}, x''_0 = -Ak^2 e^{kit}$, 于是得到 $A = 1 - 2i$, 那么解为 $x = e^{-kt}(c_1 \cos kt + c_2 \sin kt) + \sin kt - 2 \cos kt$.

$$30. 2 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + y = 4 - e^x.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 e^{-\frac{1}{2}x} + c_2 e^{-x}$, 设特解为 $y_{01} = A, y_{02} = Be^x$, 则有 $A = 4, B = -\frac{1}{6}$, 那么就有解为 $y = c_1 e^{-\frac{x}{2}} + c_2 e^{-x} + 4 - \frac{1}{6}e^x$.

$$31. 2y'' + 5y' = \cos^2 x.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 + c_2 e^{-\frac{5}{2}x}$, 因为 $\cos^2 x = \frac{1+\cos 2x}{2}$, 则我们设特解为 $y_{01} = Ax, y_{02} = Be^{2xi}$, 于是可得到 $A = \frac{1}{10}, B = \frac{-4-5i}{164}$, 那么解为 $y = c_1 + c_2 e^{-\frac{5}{2}x} + \frac{x}{10} - \frac{1}{41} \cos 2x + \frac{5}{164} \sin 2x$.

$$32. y'' + y = \sin x \cos x.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 \cos x + c_2 \sin x$, 因为 $2 \sin x \cos x = \sin 2x$, 所以设特解为 $y_0 = Ae^{2ix}$, 则有 $y_0'' = -4Ae^{2ix}$, 可得 $A = -\frac{1}{6}$, 那么解为 $y = c_1 \cos x + c_2 \sin x - \frac{1}{6} \sin 2x$.

$$33. y'' - 2y' + 2y = e^{-x} \cos x.$$

解: 齐次方程的通解为 $\tilde{y} = e^x(c_1 \cos x + c_2 \sin x)$, 设特解为 $y_0 = Ae^{(-1+i)x}$, 则有 $y_0' = A(-1+i)e^{(-1+i)x}$, $y_0'' = -2Aie^{(-1+i)x}$, 于是可得 $A = \frac{1+i}{8}$, 那么解为 $y = e^x(c_1 \cos x + c_2 \sin x) + \frac{1}{8}e^{-x}(\cos x - \sin x)$.

$$34. y'' + 4y = x \sin 2x.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 \cos 2x + c_2 \sin 2x$, 设特解为 $y_0 = x(Ax + B)e^{2ix}$, 则有 $y_0'' = [2A + 4i(2Ax + B) - 4(Ax^2 + Bx)]e^{2ix}$, 于是可得 $A = -\frac{i}{8}, B = \frac{1}{16}$, 那么解为 $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{8}x^2 \cos 2x + \frac{1}{16}x \sin 2x$.

$$35. y''' - y'' - 4y' + 4y = x^2 + 3.$$

解: 齐次方程的通解为 $\tilde{y} = c_1 e^x + c_2 e^{2x} + c_3 e^{-2x}$, 设特解为 $y_0 = Ax^2 + Bx + C$, 则有 $y_0' = 2Ax + B, y_0'' = 2A, y_0''' = 0$, 于是可得 $A = \frac{1}{4}, B = \frac{1}{2}, C = \frac{11}{8}$, 那么解为 $y = c_1 e^x + c_2 e^{2x} + c_3 e^{-2x} + \frac{1}{4}x^2 + \frac{1}{2}x + \frac{11}{8}$.

$$36. y^{(4)} + 2y'' + y = x.$$

解: 齐次方程的通解为 $\tilde{y} = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$, 设特解为 $y_0 = Ax + B$, 则有 $y_0'' = y_0^{(4)} = 0$, 于是可得 $A = 1, B = 0$, 那么解为 $y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x + x$.

37. 设 $y = (c_1 + x)e^x + c_2 e^{-2x}$ 是微分方程 $y'' + ay' + by = ge^{cx}$ 的通解, 则常数 a, b, c, g 分别等于多少?

解: 因为通解为 $y = (c_1 + x)e^x + c_2 e^{-2x}$, 则可见 $\lambda_1 = 1, \lambda_2 = -2$, 所以 $a = 1, b = -2, c = 1$, 有上式可见特解为 $y_0 = xe^x$, 则有 $y_0' = e^x + xe^x, y_0'' = 2e^x + xe^x$, 所以有 $g = 3$. 即有 $a = 1, b = -2, c = 1, g = 3$.

38. 设 $y = x \sin x$ 为 $y'' + by' + cy = A \cos x + B \sin x$ 的一个解, 则常数 b, c, A, B 分别等于多少?

解: 因为 $y = x \sin x$ 是解, 可见非齐次项 e^{aix} 中的 ai 一定是特征根 (不然不会有 x), 所以有 $\lambda_1 = i, \lambda_2 = -i$, 那么 $b = 0, c = 1$, 又有 $y' = \sin x + x \cos x, y'' = 2 \cos x - x \sin x$, 我们可以得到 $A = 2, B = 0$. 即有 $b = 0, c = 1, A = 2, B = 0$.

39. 设 $y = x^2 e^x$ 为 $y'' + by' + cy = Ae^x$ 的一个解, 则常数 b, c, A 分别等于多少?

解: $y = x^2 e^x$ 是解, 所以有非线性项可见 $\lambda = 1$ 是 2 重根, 所以 $b = -2, c = 1$, 又因为 $y' = 2xe^x + x^2 e^x, y'' = (2 + 4x + x^2)e^x$, 则有 $A = 2$. 即有 $b = -2, c = 1, A = 2$.

40. 求一个阶数尽可能低的常系数线性齐次微分方程, 使得函数 $y_1 = 2xe^x$ 与 $y_2 = 3 \sin 2x$ 是它的解.

解: 可见 y_1, y_2 是线性无关的. 则一定还有 $\cos 2x$, y_1 前面有 x , 则至少是两重的. 所以 $\lambda_1 = 2i, \lambda_2 = -2i, \lambda_3 = \lambda_4 = 1$, 所以就有 λ 应满足 $\lambda^4 - 2\lambda^3 + 5\lambda^2 - 8\lambda + 4 = 0$, 所以就方程 $y^{(4)} - 2y''' + 5y'' - 8y' + 4y = 0$.

41. 设 $f(x)$ 具有二阶连续导数, $f(1) = 0, f'(1) = 0$, 并设 $x > 0$ 时

$$(3x^2 - 2f(x))ydx - (x^2 f'(x) + \sin y)dy = 0$$

为全微分方程, 求 $f(x)$, 并求上述全微分方程的通解.

解: 有全微分知识可知: $3x^3 - 2f(x) = -x^2 f''(x) - 2x f'(x)$, 即 $f(x)$ 满足 $x^2 f''(x) + 2x f'(x) - 2f(x) + 3x^3 = 0, f(1) = 0, f'(1) = 0$, 令 $x = e^t$, 则有 $f'' + f' - 2f = -3e^{3t}$, 所以 $f(t) = c_1 e^{-2t} + c_2 e^t - \frac{3}{10} e^{3t}$, 即 $f(x) = \frac{x}{2} - \frac{1}{5x^2} - \frac{3}{10} x^3$. 所以 $(\frac{18}{5} x^3 + \frac{2}{5} x^{-2} - x)ydx - (\frac{x^2}{2} + \frac{2}{5x} - \frac{9}{10} x^4 + \sin y)dy = 0$, 则有 $\cos y + \frac{9}{10} x^4 - \frac{2}{5x} - \frac{x^2}{2} = c$.

42. 设 $f(x)$ 二阶可导, 并设 $f'(x) = f(1-x)$, 求 $f(x)$.

解: $f''(x) = (f'(x))' = -f'(1-x) = -f(x)$, 所以 $f(x) = c_1 \cos x + c_2 \sin x$, 又因为 $f'(x) = f(1-x)$, 所以有 $c_2 \cos x - c_1 \sin x = c_1 \cos(1-x) + c_2 \sin(1-x)$, 则有 $c_2 = \frac{1-\sin 1}{\cos 1}$. 所以可以得到 $f(x) = c_1(\cos x + \frac{1-\sin 1}{\cos 1})$.

43. 求 $y'' - y = e^{|x|}$ 的通解.

解: $x \geq 0$ 时: $y'' - y = e^x \Rightarrow y = (c_1 - \frac{1}{2})e^x + (c_2 + \frac{1}{2})e^{-x} + \frac{1}{2}xe^x$; $x < 0$ 时: $y'' - y = e^{-x} \Rightarrow y = c_1e^x + c_2e^{-x} - \frac{1}{2}xe^{-x}$.

44. 设 $f(x)$ 二阶可导, $f(x) + f'(\pi - x) = \sin x, f(\frac{\pi}{2}) = 0$. 求 $f(x)$.

解: 令 $x' = \pi - x$, 可得 $f'(x) = \sin x - f(\pi - x)$, 所以 $f''(x) = \cos x + f'(\pi - x) = \cos x + \sin x - f(x)$, 则有 $f(x) = c_1 \cos x + c_2 \sin x - \frac{1}{2}x \cos x + \frac{1}{2}x \sin x$. 又因为 $f(x) + f(\pi - x) = \sin x, f(\frac{\pi}{2}) = 0$, 就得到 $c_1 = \frac{\pi}{4} - \frac{1}{2}, c_2 = -\frac{\pi}{4}$. 所以有 $f(x) = (\frac{\pi}{4} - \frac{1}{2} - \frac{x}{2}) \cos x + (-\frac{\pi}{4} + \frac{x}{2})$.

45. 设 $f(x)$ 是连续函数, 并且满足 $f(x) = e^x + \int_0^x (x-t)f(t)dt$, 求 $f(x)$.

解: 因为 $f(x) = e^x + \int_0^x (x-t)f(t)dt = e^x + x \int_0^x f(t)dt - \int_0^x tf(t)dt$ 所以 $f'(x) = e^x + \int_0^x f(t)dt + xf(x) - xf(x)$, $f''(x) = e^x + f(x)$. 所以就有 $f(x) = c_1e^x + c_2e^{-x} + \frac{1}{2}xe^x$. 又因为 $f(0) = 1, f'(0) = 1$, 所以 $c_1 = \frac{3}{4}, c_2 = \frac{1}{4}$. 所以 $f(x) = \frac{3}{4}e^x + \frac{1}{4}e^{-x} + \frac{1}{2}xe^x$.

46. 设二阶常系数线性微分方程 $y'' + \alpha y' + \beta y = \gamma e^x$ 的一个特解为 $y = e^{2x} + (1+x)e^x$, 是确定常数 α, β, γ , 并求该方程的通解.

解: 将特解代入得
$$\begin{cases} 4 + 2\alpha + \beta = 0 \\ 3 + 2\alpha + \beta = \gamma \\ 1 + \alpha + \beta = 0 \end{cases} \Rightarrow \begin{cases} \alpha = -3 \\ \beta = 2 \\ \gamma = -1 \end{cases}$$
 所以就可以得到 $y = c_1e^x + c_2e^{2x} + xe^x$.

47. 质量为 1 克的质点被一力从某中心沿直线推开, 该力的大小与这个中心到质点的距离成正比 (比例常数为 4); 介质阻力和运动速度成正比 (比例常数为 3). 在运动开始时, 质点于中心的距离为 1 厘米, 速度为 0, 求质点的运动方程.

解: 设距离为 x , 则 $x''(t) = 4x(t) - 3x'(t), x(0) = 1, x'(0) = 0$, 所以就有 $x(t) = c_1e^{-4t} + c_2e^t, c_1 = \frac{1}{5}, c_2 = \frac{4}{5}$, 所以就有 $x(t) = \frac{1}{5}(4e^t + e^{-4t})$.

48. 重量为 P 牛顿的列车沿水平轨道作直线运动, 当速度不大使列车受到阻力 $R = (a + bv)P$ 牛顿, 其中 a, b 是常数, v 是列车速度. 设机车的牵引力是 F 牛顿. 当 $t = 0$ 时, $s = 0$ (s 为走过的路程), $v = 0$. 求火车的运动方程.

解: $S'' = F/(P/g) - (a + bS')P/(P/g) = (Fg/P - ag) - bgS', S(0) = 0, S'(0) = 0$

$S(t) = c_1e^{-bgt} + c_2 + \frac{1}{bg}(Fg/P - ag)t \Rightarrow c_1 = -c_2 = \frac{1}{b^2g}(F/g - a)$,

所以有 $S(t) = \frac{1}{b^2g}(F/g - a)(e^{-bgt} - 1 + bgt)$.

49. 一电阻 $R = 250$ 欧, 电感 $L = 1$ 亨, 电容 $C = 10^{-4}$ 法的串联电路, 外加直流电压 $E = 100$ 伏, 当时间 $t = 0$ 时, 电流 $i = 0$, $\frac{di}{dt} = 100$ 安 / 秒, 求电路重点流域时间的函数关系.

解: $100 = 250i + \int_0^t idt/c + li \Rightarrow i'' + 250i' + 10^4i(t) = 0 \Rightarrow i(t) = c_1e^{-50t} + c_2e^{-200t}, i(0) = 0, i'(0) = 100$, 则有 $c_1 = -c_2 = \frac{2}{3} \Rightarrow i(t) = \frac{2}{3}(e^{-50t} - e^{-200t})$.

求下列方程的通解 (50~52):

50. $x^2y'' + 3xy' + y = 0$.

解: 令 $x = e^t$, 那么原式就变为: $y''(t) + 2y'(t) + y = 0$. 该式的通解为 $y = c_1e^t + c_2te^t$, 那么就得到 $y = \frac{1}{x}(c_1 + c_2 \ln x), x > 0$.

51. $x^2y'' - 2xy' + 2y = x \ln x$. (题目要改一下)

解: 令 $x = e^t$, 有 $y''(t) - 3y'(t) + 2y = te^t$. 该式的通解为 $y = c_1e^t + c_2e^{2t} + (At^2 + Bt)e^t$, 有 $A = -\frac{1}{2}, B = -1$, 那么就得到解为 $y = x[c_1 - \frac{1}{2}(\ln x)^2 - \ln x] + c_2x^2, x > 0$.

52. $x^3y''' - 3x^2y'' + 6xy' - 6y = 0$.

解: $\frac{d^3y}{dx^3} = \frac{1}{x^3}(\frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 2\frac{dy}{dt})$, 那么就有 $y''' - 6y'' + 11y' - 6y = 0$, 通解为 $y = c_1e^t + c_2e^{2t} + c_3e^{3t}$, 于是有解为 $y = c_1x + c_2x^2 + c_3x^3$.

53. 设 $r = \sqrt{x^2 + y^2 + z^2} > 0$, $f(r)$ 具有二阶导数, 且满足 $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$. 求 $f(r)$.

解:

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= r^{-2} x^2 \frac{\partial^2 f}{\partial r^2} + (r^{-1} - x^2 r^3) \frac{\partial f}{\partial r} \\ \Rightarrow \frac{\partial^2 f}{\partial r^2} + 2r^{-1} \frac{\partial f}{\partial r} &= 0 \\ \Rightarrow r^2 \frac{\partial^2 f}{\partial r^2} + 2r \frac{\partial f}{\partial r} &= 0 \\ f(r) &= c_1 + \frac{c_2}{r}\end{aligned}$$

已知下列方程对应的齐次线性方程的一个解 y_1 , 求该方程的通解 (54~ 56):

54. $x^3 y''' - xy' + y = 0$, 已知 $y_1 = x$.

解: $y'' - x^{-2}y' + x^{-3}y = 0$, 令 $y = xu$,

$$\begin{aligned}\Rightarrow xu'' + [2 - \frac{1}{x}]u' &= 0 \\ \Rightarrow u &= c_1 + c_2 e^{-\frac{1}{x}} \\ y &= c_1 x + c_2 x e^{-\frac{1}{x}}\end{aligned}$$

55. $(1 - x^2)y''' - xy'' + y' = 0$, 已知 $y_1 = x^2$.

解: 令 $z = y'$, 则 $z = 2x$. 原式就变为 $z'' - \frac{x}{1-x^2}z' + \frac{x}{1-x^2}z = 0$

$$\begin{aligned}\Rightarrow 2xu'' + [4 - \frac{2x^2}{1-x^2}]u' &= 0 \\ z &= c_1 x + c_2 \int \frac{1}{x^2 \sqrt{1-x^2}} \\ z &= c_1 x + c_2 \sqrt{1-x^2} \\ y &= c_1 + c_2 x^2 + c_3 (x\sqrt{1-x^2} + \arcsin x)\end{aligned}$$

56. $(1 - x^2)y'' + 2xy' - 2y = -2$, 已知 $y_1 = x$.

解: $y'' + \frac{2x}{1-x^2}y' - \frac{2}{1-x^2}y = 0$, 令 $y = xu$

$$\begin{aligned}\Rightarrow xu'' + [2 + \frac{2x}{1-x^2}]u' &= 0 \\ \Rightarrow u &= c_1 + c_2 (x + \frac{1}{x}) \\ \Rightarrow y &= c_1 x + c_2 (x^2 + 1) + y^*, y^* = 1 \\ \Rightarrow y &= c_1 x + c_2 (x^2 + 1) + 1\end{aligned}$$

57. 设 $y_1(x)$ 和 $y_2(x)$ 是二阶非齐次线性方程

$$y'' + p(x)y' + q(x)y = f(x)$$

所对应的齐次线性方程两个线性无关的解, 它们的朗斯基行列式为 $W(x)$. 试证明: 该非齐次线性方程的通解为

$$y = c_1 y_1(x) + c_2 y_2(x) + \int_{x_0}^x \frac{1}{W(\xi)} [y_1(\xi)y_2(x) - y_2(\xi)y_1(x)] f(\xi) d\xi$$

其中 $p(x), q(x)$ 和 $f(x)$ 在区间 (a, b) 内连续, $x_0 \in (a, b)$.

解: 变动任意常数法。 $y = u_1 y_1 + u_2 y_2$

$$\Rightarrow \begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = f(x) \end{cases}$$

$$\Rightarrow u'_1 = -\frac{y_2 f}{w}, u'_2 = \frac{y_1 f}{w}$$

$$y = c_1 y_1(x) + c_2 y_2(x) + \int_{x_0}^x \frac{1}{w(\xi)} [y_1(\xi) y_2(x) - y_2(\xi) y_1(x)] f(\xi) d\xi$$

求下列方程的通解 (58~ 61):

$$58. y'' - 2y' + y = \frac{e^x}{x}.$$

解: $y = c_1 e^x + c_2 x e^x + y^*, y^* = e^x x \ln |x|$ 代入上题的公式得到

$$y = e^x (x \ln |x| + c_1 + c_2 x)$$

$$59. y'' - y = 2 \sec^3 x.$$

解:

$$y = c_1 \cos x + c_2 \sin x + y^*, y^* = -\frac{\cos 2x}{\cos x}$$

$$\Rightarrow y = c_1 \cos x + c_2 \sin x - \frac{\cos 2x}{\cos x}$$

$$60. y''' + 4y' = 4 \cot 2x.$$

解:

$$y = c_1 \sin 2x + c_2 \cos 2x + c_3 + y^*$$

$$(y')^* = \int (\cos 2\xi \sin 2x - \sin 2\xi \cos 2x) \ln |\cos 2\xi| d\xi$$

$$y = c_1 \sin 2x + c_2 \cos 2x + c_3 + \frac{1}{2} \ln |\sin 2x| - \frac{1}{2} \cos 2x \ln |\cos 2x - \cot 2x|$$

$$61. y'' + 3y' + 2y = \frac{1}{e^x + 1}.$$

解: $y = c_1 e^{-2x} + c_2 e^{-x} + y^*, y^* = (e^{-x} + e^{-2x}) \ln(e^x + 1) + e^{-2x}(-e^x + 1)$

$$\Rightarrow y = c_1 e^{-2x} + c_2 e^{-x} + (e^{-x} + e^{-2x}) \ln(e^x + 1)$$

62. 用幂级数解法求 $y'' + 4xy = 0$ 的通解。

解: 令 $y = \sum_{i=0}^{\infty} a_i x^i$,

$$y'' = \sum_{i=2}^{\infty} a_i x^{i-2} i(i-1) = \sum_{i=0}^{\infty} (i+1)(i+2) a_{i+2} x^i$$

$$\Rightarrow \sum_{i=1}^{\infty} [(i+1)(i+2) a_{i+2} + 4a_{i-1}] x^i + 2a_2 = 0$$

$$\Rightarrow a_{i+2} = a_{i-1} \frac{4}{(i+1)(i+2)}, a_2 = 0$$

$$a_{3k} = \frac{(-4)^k a_0}{3k(3k-1) \cdots 6 \cdot 5 \cdot 3 \cdot 2}$$

$$a_{3k+1} = \frac{(-4)^k a_0}{(3k+1)3k \cdots 7 \cdot 6 \cdot 4 \cdot 3}$$

$$a_{3k+2} = 0$$

$$\Rightarrow y = c_1 (1 + \cdots + \frac{(-4)^k a_0 x^{3k}}{3k(3k-1) \cdots 6 \cdot 5 \cdot 3 \cdot 2} + \cdots) + c_2 (x + \cdots + \frac{(-4)^k a_0 x^{3k+1}}{(3k+1)3k \cdots 7 \cdot 6 \cdot 4 \cdot 3})$$

63. 用广义幂级数法求 $4xy'' + 2y' + y = 0$ 的通解。

解: 令 $y = \sum_{i=0}^{\infty} a_i x^i + \sum_{i=0}^{\infty} b_i x^{i+\frac{1}{2}}$

$$y' = \sum_{i=0}^{\infty} (i+1)(i+2) a_{i+2} x^i + \sum_{i=0}^{\infty} (i+\frac{1}{2})(i-\frac{1}{2}) b_{i+2} x^{i-\frac{3}{2}}$$

$$\begin{aligned}
&\Rightarrow \sum_{i=1}^{\infty} [4i(i+1)a_{i+1} + 2(i+1)a_{i+1} + a_i]x^i + (2a_1 + a_0) \\
&+ \sum_{i=0}^{\infty} [4(i+\frac{3}{2})(i+\frac{1}{2})b_{i+1} + 2(i+\frac{3}{2})b_{i+1} + b_i]x^{i+\frac{1}{2}} = 0 \\
a_1 &= -\frac{a_0}{2}, a_{i+1} = -\frac{a_i}{(2i+2)(2i+1)}, b_{i+1} = -\frac{b_i}{(2i+2)(2i+3)} \\
y &= c_1(1 - \frac{x}{2!} + \frac{x^2}{4!} - \cdots + (-1)^k \frac{x^k}{(2k)!} + \cdots) \\
&+ c_2(x^{\frac{1}{2}} - \frac{1}{3!}x^{\frac{3}{2}} + \frac{1}{5!}x^{\frac{5}{2}} - \cdots + (-1)^{k-1} \frac{x^{\frac{2k-1}{2}}}{(2k-1)!}) \\
&= c_1 \cos \sqrt{x} + c_2 \sin \sqrt{x}
\end{aligned}$$

64. 用幂级数解法求 $(1-x^2)y'' - xy' + \frac{1}{9}y = 0$ 满足 $y(0) = \sqrt{3}/2$, $y'(0) = \frac{1}{6}$ 的解。

解: 同理 $\Rightarrow \sum_{i=2}^{\infty} [(i+1)(i+2)a_{i+2} - i(i-1)a_i - ia_i + \frac{1}{9}a_i]x^i + \frac{a_0}{9} + a_2 + x(\frac{a_1}{9} - a_1 + a_3) = 0$

$$\Rightarrow a_{i+2} = (i^2 - \frac{1}{9}) \frac{a_i}{(i+1)(i+2)}, a_2 = -\frac{a_0}{9}, a_3 = \frac{8}{9}a_1$$

由 $y(0) = \frac{\sqrt{3}}{2}, y'(0) = \frac{1}{6} \Rightarrow a_0 = \frac{\sqrt{3}}{2}, a_1 = \frac{1}{6}$

$$\begin{aligned}
y &= \frac{\sqrt{3}}{2} [1 - \frac{1}{9} \frac{x^2}{2!} - \frac{1}{9} (2^2 - \frac{1}{9}) \frac{x^4}{4!} - \cdots - \frac{1}{9} (2^2 - \frac{1}{9}) \cdots [(2k-2)^2 - \frac{1}{9}] \frac{x^{2k}}{(2k)!} - \cdots] \\
&+ \frac{1}{6} [x + (1 - \frac{1}{9})(3^2 - \frac{1}{9}) \frac{x^5}{5!} + \cdots + (1 - \frac{1}{9})(3^2 - \frac{1}{9}) \cdots [(2k-2)^2 - \frac{1}{9}] \frac{x^{2k+1}}{(2k+1)!} + \cdots]
\end{aligned}$$

65. 对于什么样的常数 p 和 q , 方程 $\frac{d^2x}{dt^2} + p\frac{dx}{dt} + qx = 0$ 的所有解当 $t \rightarrow +\infty$ 时趋于零?

解: 当 $\Delta > 0$ 时, $\lambda_1 = -\frac{p}{2} + \frac{\sqrt{\Delta}}{2}, \lambda_2 = -\frac{p}{2} - \frac{\sqrt{\Delta}}{2}$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} c_1 e^{\lambda_1 t} + c_2 e^{-\lambda_2 t} = 0$$

$$\Leftrightarrow \lambda_2 < \lambda_1 < 0 \Rightarrow p > 0, q > 0$$

当 $\Delta = 0$ 时, 同理可得 $\Leftrightarrow \lambda_1 = \lambda_2 < 0 \Rightarrow p > 0, q > 0$

当 $\Delta < 0$ 时, $\lambda_1 = \alpha + i\beta, \lambda_2 = \alpha - i\beta, \alpha = -\frac{p}{2}$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} e^{\alpha t} (c_1 \sin \beta t + c_2 \cos \beta t) = 0$$

$$\Leftrightarrow \alpha < 0 \Rightarrow p > 0, q > 0$$

综合得到 $p > 0, q > 0$

66. 给定方程 $\frac{d^2x}{dt^2} + p\frac{dx}{dt} + qx = f(t)$, 其中常数 $p > 0, q > 0$, 函数 $f(t)$ 在 $0 \leq t < +\infty$ 上连续。试证明: (1) 如果 $f(t)$ 在 $0 \leq t < +\infty$ 上有界, 则上述方程的每一个解在 $0 \leq t < +\infty$ 上也有界; (2) 如果当 $t \rightarrow +\infty$ 时 $f(t) \rightarrow 0$, 则上述方程的每一个解当 $t \rightarrow +\infty$ 时都趋于零。解: 利用 57 与 65 的结论, 可知对于齐次通解 (1)、(2) 显然成立, 因此只需要证明特解即可。

当 $\Delta > 0$ 时

$$(1) \lambda_2 < \lambda_1 < 0, y_1 = e^{\lambda_1 t}, y_2 = e^{\lambda_2 t}$$

$$W(\xi) = (\lambda_2 - \lambda_1) e^{\lambda_1 + \lambda_2 \xi}$$

$$\begin{aligned}
|y *| &= \left| \int_0^t \frac{1}{\lambda_2 - \lambda_1} e^{-(\lambda_1 + \lambda_2)\xi} (e^{\lambda_1 \xi + \lambda_2 t} - e^{\lambda_2 \xi + \lambda_1 t}) f(\xi) d\xi \right| \\
&\leq \frac{|f|_{\max}}{\lambda_1 - \lambda_2} \int_0^t |e^{\lambda_2(t-\xi)} - e^{\lambda_1(t-\xi)}| d\xi
\end{aligned}$$

$$\begin{aligned}
&= \frac{|f|_{\max}}{\lambda_1 - \lambda_2} - \frac{1}{\lambda_1} [1 - e^{\lambda_1 t}] + \frac{1}{\lambda_2} (1 - e^{\lambda_2 t}) \\
&\leq \frac{|f|_{\max}}{\lambda_1 - \lambda_2} \left[-\frac{1}{\lambda_1} \right]
\end{aligned}$$

所以有界。

$$(2) \forall \epsilon > 0, \exists A > 0,$$

$$s.t. \forall t > A, |f(t)| < \epsilon, e^{\lambda_1 t} < \epsilon, e^{\lambda_2 t} < \epsilon, \forall t, \exists M > 0, |f(t)| \leq M,$$

$$\begin{aligned}
|y^*| &\leq \left| \int_0^A y(\xi) d\xi \right| + \left| \int_A^t y(\xi) d\xi \right| \\
&\leq \frac{M}{\lambda_1 - \lambda_2} \left[-\frac{1}{\lambda_1} (e^{\lambda_1(t-A)} - e^{\lambda_1 t}) + \frac{1}{\lambda_2} (e^{\lambda_2(t-A)} - e^{\lambda_2 t}) \right] + \frac{\epsilon}{\lambda_1 - \lambda_2} \left[-\frac{1}{\lambda_1} \right]
\end{aligned}$$

只需要 $t > 2A$ 就有

$$\leq \frac{M}{\lambda_1 - \lambda_2} \left(-\frac{2\epsilon}{\lambda_1} - \frac{2\epsilon}{\lambda_2} \right) + \frac{\epsilon}{\lambda_1 - \lambda_2} \left(-\frac{1}{\lambda_1} \right) \leq c_0 \epsilon (t > 2A)$$

当 $\Delta = 0$ 时

$$(1) \lambda_1 = \lambda_2 < 0, y_1 = e^{\lambda_1 t}, y_2 = te^{\lambda_1 t}, W(\xi) = e^{2\lambda_1 \xi}$$

$$\begin{aligned}
|y^*| &= \left| \int_0^t (t - \xi) e^{\lambda_1(t-\xi)} f(\xi) d\xi \right| \\
&\leq |f|_{\max} \left| \left[-\frac{t - \xi}{\lambda_1} e^{\lambda_1(t-\xi)} + \frac{1}{\lambda_1^2} e^{\lambda_1(t-\xi)} \right] \Big|_0^t \right| \\
&= |f|_{\max} \frac{1}{\lambda_1^2} |1 - e^{\lambda_1 t} + \lambda_1 t e^{\lambda_1 t}|
\end{aligned}$$

所以有界。

$$(2) \text{ 同理, } \forall \epsilon > 0, \exists A, s.t. \forall t > A, |f(t)| < \epsilon$$

$$\begin{aligned}
&e^{\lambda_1 t} < \epsilon, te^{\lambda_1 t} < \epsilon, \forall t, |f(t)| < M \\
|y^*| &\leq \left| \int_0^A y(\xi) d\xi \right| + \left| \int_A^t y(\xi) d\xi \right| \\
&M \left| \frac{-(t-A)}{\lambda_1} e^{\lambda_1(t-A)} \right| + M \left| \frac{t}{\lambda_1} e^{\lambda_1 t} \right| + \epsilon \frac{1}{\lambda_1^2} |1 - e^{\lambda_1 t} + \lambda_1 t e^{\lambda_1 t}| < c_0 \epsilon
\end{aligned}$$

只需要 $t > 2A$ 即可。

当 $\Delta < 0$ 时

$$(1) \lambda_1 = \alpha + i\beta, \lambda_2 = \alpha - i\beta, \alpha = -\frac{\rho}{2}$$

$$W(\xi) = e^{2\alpha \xi}, y_1 = e^{\alpha t} \cos \pi t, y_2 = e^{\alpha t} \sin \pi t$$

$$\begin{aligned}
|y^*| &= \left| \int_0^t e^{\alpha(t-\xi)} \sin \beta(t - \xi) f(\xi) d\xi \right| \\
&\leq |f|_{\max} \int_0^t e^{\alpha(t-\xi)} d\xi \\
&= |f|_{\max} \frac{1}{(-\alpha)} (1 - e^{\alpha t})
\end{aligned}$$

所以有界。

$$(2) \text{ 同理 } \forall \epsilon > 0, \exists A, s.t. \forall t > A, |f(t)| < \epsilon, e^{\alpha t} < \epsilon, \forall t, |f(t)| < M$$

$$\begin{aligned}
|y^*| &\leq \left| \int_0^A y(\xi) d\xi \right| + \left| \int_A^t y(\xi) d\xi \right| \\
&\leq M \frac{1}{(-\alpha)} |e^{\alpha(t-A)} - e^{\alpha t}| + \epsilon \frac{1}{-\alpha} [1 - e^{\alpha(t-A)}] \leq c_0 \epsilon
\end{aligned}$$

只需要 $t > 2A$ 即可。

第三章

1. 将第二章习题 1 ~ 3 中的“函数”改成“向量函数”，则这些命题仍成立，试叙述并证明这些命题。

2. 对于非齐次线性方程组，叙述并证明与第二章习题 4 相应的习题。

3. 设 $X(t)$ 是齐次线性方程组 $\frac{dx}{dt} = A(t)x$ 的一个基本解矩阵， $A(t)$ 在区间 (a, b) 内连续， $W(t)$ 是 $X(t)$ 的朗斯基行列式。试证明下述刘维尔公式：

$$W(t) = W(t_0)e^{\int_{t_0}^t \sum_{i=1}^n a_{ii}(\tau) d\tau}, t_0 \in (a, b), t \in (a, b).$$

并证明：如果所给的齐次线性方程组是由高阶齐次线性方程经变换 (3.3) 得到的，则上述刘维尔公式与第二章习题 5 的刘维尔公式一致。

1, 2, 3 证明同第二章，略。

4. 设 $x_1(t)$ 和 $x_2(t)$ 分别是 $\frac{dx}{dt} - A(t)x = f_1(t)$ 和 $\frac{dx}{dt} - A(t)x = f_2(t)$ 的解，试证明 $x_1 + x_2(t)$ 是 $\frac{dx}{dt} - A(t)x = f_1(t) + f_2(t)$ 的解。

证明： $\frac{d(x_1(t) + x_2(t))}{dt} - A(t)(x_1(t) + x_2(t)) = \frac{dx_1}{dt} - A(t)x_1 + \frac{dx_2}{dt} - A(t)x_2 = f_1(t) + f_2(t)$ 。

5. 设 $A(t)$ 是实矩阵， t 是实变量， $x(t) = u(t) + iv(t)$ 是方程 $\frac{dx}{dt} - A(t)x = \varphi(t) + i\psi(t)$ 的解，其中 $u(t)$ ， $v(t)$ ， $\varphi(t)$ ， $\psi(t)$ 都是实函数， $i = \sqrt{-1}$ 是虚单位。试证明 $u(t)$ 和 $v(t)$ 分别满足 $\frac{du(t)}{dt} - A(t)u(t) = \varphi(t)$ 和 $\frac{dv(t)}{dt} - A(t)v(t) = \psi(t)$ 。

证明：将 $x(t) = u(t) + iv(t)$ 代入 $\frac{dx}{dt} - A(t)x = \varphi(t) + i\psi(t)$

$$\Rightarrow \frac{du}{dt} - A(t)u(t) + (\frac{dv}{dt} - A(t)v(t))i = \varphi(t) + i\psi(t)$$

$$\Rightarrow \begin{cases} \frac{du}{dt} - A(t)u(t) = \varphi(t) \\ \frac{dv}{dt} - A(t)v(t) = \psi(t) \end{cases}$$

$\Rightarrow u(t)$ ， $v(t)$ 分别是 $\frac{du(t)}{dt} - A(t)u(t) = \varphi(t)$ ， $\frac{dv(t)}{dt} - A(t)v(t) = \psi(t)$ 的解。

求下列方程组的解 (6 ~ 20)：

$$6. \begin{cases} \frac{dx}{dt} = y - 3x, \\ \frac{dy}{dt} = 8x - y. \end{cases}$$

解： $A = \begin{pmatrix} -3 & 1 \\ 8 & -1 \end{pmatrix}$

$$\Rightarrow |A - \lambda E| = \begin{vmatrix} -3-\lambda & 1 \\ 8 & -1-\lambda \end{vmatrix} = (3+\lambda)(1+\lambda) - 8 = (\lambda+5)(\lambda-1) =$$

$$0 \Rightarrow \lambda_1 = -5, \quad \lambda_2 = 1$$

$$v_1 = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \Rightarrow \begin{pmatrix} 2 & 1 \\ 8 & 4 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = 0 \Rightarrow \begin{cases} 2\alpha_1 + \beta_1 = 0 \\ 8\alpha_1 + 4\beta_1 = 0 \end{cases} \Rightarrow v_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} \Rightarrow \begin{pmatrix} -4 & 1 \\ 8 & -2 \end{pmatrix} \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} = 0 \Rightarrow \begin{cases} -4\alpha_2 + \beta_2 = 0 \\ 8\alpha_2 - 2\beta_2 = 0 \end{cases} \Rightarrow$$

$$v_2 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = c_1 e^{-5t} \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 e^x \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

$$7. \begin{cases} \frac{dx}{dt} = x - y, \\ \frac{dy}{dt} = x + y. \end{cases}$$

$$\text{解: } A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad |A - \lambda E| = \begin{vmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 + 1 =$$

$$2 - 2\lambda + \lambda^2 = 0 \Rightarrow \lambda_1 = 1 + i, \quad \lambda_2 = 1 - i.$$

$$v_1 = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \Rightarrow \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = 0 \Rightarrow \begin{cases} -i\alpha_1 - \beta_1 = 0 \\ \alpha_1 - i\beta_1 = 0 \end{cases} \Rightarrow v_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$e^t(\cos t + i \sin t) \begin{pmatrix} 1 \\ -i \end{pmatrix} = e^t \begin{pmatrix} \cos t + i \sin t \\ -i \cos t + \sin t \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = e^t \left(c_1 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + c_2 \begin{pmatrix} \sin t \\ -\cos t \end{pmatrix} \right)$$

$$8. \begin{cases} \frac{dx}{dt} = x - 5y, \\ \frac{dy}{dt} = 2x - y. \end{cases}$$

$$\text{解: } A = \begin{pmatrix} 1 & -5 \\ 2 & -1 \end{pmatrix}, \quad |A - \lambda E| = \begin{vmatrix} 1-\lambda & -5 \\ 2 & -1-\lambda \end{vmatrix} = -(1-\lambda^2) + 10 =$$

$$\lambda^2 + 9 = 0 \Rightarrow \lambda = \pm 3i.$$

$$v = \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1-3i & -5 \\ 2 & -1-3i \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} = 0 \Rightarrow \begin{cases} (1-3i)\alpha_1 - 5\beta_1 = 0 \\ 2\alpha_1 - (1+3i)\beta_1 = 0 \end{cases} \Rightarrow$$

$$v = \begin{pmatrix} 5 \\ 1-3i \end{pmatrix}.$$

$$\begin{aligned}
& (\cos 3t + i \sin 3t) \begin{pmatrix} 5 \\ 1 - 3i \end{pmatrix} = \begin{pmatrix} 5 \cos 3t + 5i \sin 3t \\ \cos 3t + 3 \sin 3t + (\sin 3t - 3 \cos 3t)i \end{pmatrix} \\
\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} &= c_1 \begin{pmatrix} 5 \cos 3t \\ \cos 3t + 3 \sin 3t \end{pmatrix} + c_2 \begin{pmatrix} 5 \sin 3t \\ \sin 3t - 3 \cos 3t \end{pmatrix}.
\end{aligned}$$

$$9. \begin{cases} \frac{dx}{dt} = y, \\ \frac{dy}{dt} = z, \\ \frac{dz}{dt} = x. \end{cases}$$

$$\text{解: } A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad |A - \lambda E| = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & 0 & -\lambda \end{vmatrix} = -\lambda^3 + 1 =$$

$$-(\lambda - 1)(\lambda^2 + \lambda + 1) = 0 \Rightarrow \lambda_1 = 1, \quad \lambda_2 = \frac{-1 + \sqrt{3}i}{2}, \quad \lambda_3 = \frac{-1 - \sqrt{3}i}{2}$$

$$v_1 = \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} \Rightarrow \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = 0 \Rightarrow \begin{cases} -\alpha_1 + \beta_1 = 0 \\ -\beta_1 + \gamma_1 = 0 \\ \alpha_1 - \gamma_1 = 0 \end{cases} \Rightarrow$$

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

$$v_2 = \begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{1 - \sqrt{3}i}{2} & 1 & 0 \\ 0 & \frac{1 - \sqrt{3}i}{2} & 1 \\ 1 & 0 & \frac{1 - \sqrt{3}i}{2} \end{pmatrix} \begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \end{pmatrix} = 0 \Rightarrow \begin{cases} \frac{1 - \sqrt{3}i}{2} \alpha_2 + \beta_2 = 0 \\ \frac{1 - \sqrt{3}i}{2} \beta_2 + \gamma_2 = 0 \\ \alpha_2 + \frac{1 - \sqrt{3}i}{2} \gamma_2 = 0 \end{cases} \Rightarrow$$

$$v_2 = \begin{pmatrix} 1 \\ \frac{-1 + \sqrt{3}i}{2} \\ \frac{-1 - \sqrt{3}i}{2} \end{pmatrix}.$$

$$e^{-\frac{1}{2}t} \left(\cos \frac{\sqrt{3}}{2}t + i \sin \frac{\sqrt{3}}{2}t \right) \begin{pmatrix} 1 \\ -\frac{1}{2} + \frac{\sqrt{3}}{2}i \\ -\frac{1}{2} - \frac{\sqrt{3}}{2}i \end{pmatrix}$$

$$= e^{-\frac{1}{2}t} \begin{pmatrix} \cos \frac{\sqrt{3}}{2}t + i \sin \frac{\sqrt{3}}{2}t \\ -\frac{1}{2} \cos \frac{\sqrt{3}}{2}t - \frac{\sqrt{3}}{2} \sin \frac{\sqrt{3}}{2}t + \left(\frac{\sqrt{3}}{2} \cos \frac{\sqrt{3}}{2}t - \frac{1}{2} \sin \frac{\sqrt{3}}{2}t \right) i \\ -\frac{1}{2} \cos \frac{\sqrt{3}}{2}t + \frac{\sqrt{3}}{2} \sin \frac{\sqrt{3}}{2}t + \left(-\frac{\sqrt{3}}{2} \cos \frac{\sqrt{3}}{2}t - \frac{1}{2} \sin \frac{\sqrt{3}}{2}t \right) i \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} \cos \frac{\sqrt{3}}{2} t \\ -\frac{1}{2} \cos \frac{\sqrt{3}}{2} t - \frac{\sqrt{3}}{2} \sin \frac{\sqrt{3}}{2} t \\ -\frac{1}{2} \cos \frac{\sqrt{3}}{2} t + \frac{\sqrt{3}}{2} \sin \frac{\sqrt{3}}{2} t \end{pmatrix} e^{-\frac{t}{2}} + c_3 \begin{pmatrix} \sin \frac{\sqrt{3}}{2} t \\ \frac{\sqrt{3}}{2} \cos \frac{\sqrt{3}}{2} t - \frac{1}{2} \sin \frac{\sqrt{3}}{2} t \\ -\frac{\sqrt{3}}{2} \cos \frac{\sqrt{3}}{2} t - \frac{1}{2} \sin \frac{\sqrt{3}}{2} t \end{pmatrix} e^{-\frac{t}{2}}.$$

$$10. \begin{cases} \frac{dx}{dt} = x - y, \\ \frac{dy}{dt} = x + 3y. \end{cases}$$

$$\text{解: } A = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \Rightarrow |A - \lambda E| = \begin{vmatrix} 1-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) + 1 =$$

$$\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0 \Rightarrow \lambda = 2 (\text{二重})$$

$$\Rightarrow \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0$$

$$v_1^{(1)} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v_2^{(1)} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, v_1^{(2)} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix}, v_2^{(2)} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix}_1 = \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right) e^{2t} = \begin{pmatrix} 1-t \\ t \end{pmatrix} e^{2t},$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_2 = \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right) e^{2t} = \begin{pmatrix} -t \\ 1+t \end{pmatrix} e^{2t}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = c_1 \begin{pmatrix} 1-t \\ t \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -t \\ 1+t \end{pmatrix} e^{2t}.$$

$$11. \begin{cases} \frac{dx}{dt} = y + z, \\ \frac{dy}{dt} = z + x, \\ \frac{dz}{dt} = x + y. \end{cases}$$

$$\text{解: } A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\Rightarrow |A - \lambda E| = \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -\lambda^3 + 1 + 1 + \lambda + \lambda + \lambda = -\lambda^3 + 3\lambda + 2 = 0$$

$$\Rightarrow (\lambda + 1)^2(\lambda - 2) = 0 \Rightarrow \lambda_1 = -1 (\text{二重}), \lambda_2 = 2.$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = 0$$

$$\begin{aligned}
&\Rightarrow \alpha_1 + \beta_1 + \gamma_1 = 0 \\
&\Rightarrow v_1^{(1)} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad v_2^{(1)} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \quad v_1^{(2)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad v_2^{(2)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \\
&\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} \Rightarrow \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = 0 \Rightarrow \begin{cases} -2\alpha + \beta + \gamma = 0 \\ \alpha - 2\beta + \gamma = 0 \\ \alpha + \beta - 2\gamma = 0 \end{cases} \Rightarrow \\
&v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (c_1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix})e^{-t} + c_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{2t}. \\
&12. \begin{cases} \frac{dx}{dt} = x + y - z, \\ \frac{dy}{dt} = -x + y + z, \\ \frac{dz}{dt} = x - y + z. \end{cases}
\end{aligned}$$

$$\begin{aligned}
\text{解: } A &= \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}, \quad |A - \lambda E| = \begin{vmatrix} 1-\lambda & 1 & -1 \\ -1 & 1-\lambda & 1 \\ 1 & -1 & 1-\lambda \end{vmatrix} = \\
&(1-\lambda)^3 + 1 - 1 + (1-\lambda) + (1-\lambda) + (1-\lambda) = (1-\lambda)(\lambda^2 - 2\lambda + 4)
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \lambda_1 = 1, \quad \lambda_2 = 1 + \sqrt{3}i, \quad 1 - \sqrt{3}i. \\
&v_1 = \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = 0 \Rightarrow v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}
\end{aligned}$$

$$v_2 = \begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \end{pmatrix} \Rightarrow \begin{pmatrix} -\sqrt{3}i & 1 & -1 \\ -1 & -\sqrt{3}i & 1 \\ 1 & -1 & -\sqrt{3}i \end{pmatrix} \begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \end{pmatrix} = 0 \Rightarrow v_2 = \begin{pmatrix} 1 \\ -\frac{1}{2} + \frac{\sqrt{3}i}{2} \\ -\frac{1}{2} - \frac{\sqrt{3}i}{2} \end{pmatrix}$$

$$\begin{aligned}
&(\cos \sqrt{3}t + \sin \sqrt{3}ti) \begin{pmatrix} 1 \\ -\frac{1}{2} + \frac{\sqrt{3}i}{2} \\ -\frac{1}{2} - \frac{\sqrt{3}i}{2} \end{pmatrix} = \begin{pmatrix} \cos \sqrt{3}t + \sin \sqrt{3}ti \\ (-\frac{1}{2} \cos \sqrt{3}t - \frac{\sqrt{3}}{2} \sin \sqrt{3}t) + (-\frac{1}{2} \sin \sqrt{3}t + \frac{\sqrt{3}}{2} \cos \sqrt{3}t)i \\ (-\frac{1}{2} \cos \sqrt{3}t + \frac{\sqrt{3}}{2} \sin \sqrt{3}t) + (-\frac{1}{2} \sin \sqrt{3}t - \frac{\sqrt{3}}{2} \cos \sqrt{3}t)i \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^t + (c_2 \begin{pmatrix} \cos \sqrt{3}t \\ -\frac{1}{2} \cos \sqrt{3}t - \frac{\sqrt{3}}{2} \sin \sqrt{3}t \\ -\frac{1}{2} \cos \sqrt{3}t + \frac{\sqrt{3}}{2} \sin \sqrt{3}t \end{pmatrix} + c_3 \begin{pmatrix} \sin \sqrt{3}t \\ -\frac{1}{2} \sin \sqrt{3}t + \frac{\sqrt{3}}{2} \cos \sqrt{3}t \\ -\frac{1}{2} \sin \sqrt{3}t - \frac{\sqrt{3}}{2} \cos \sqrt{3}t \end{pmatrix}) e^t
\end{aligned}$$

$$13. \begin{cases} \frac{dx}{dt} - \frac{dy}{dt} - \frac{dz}{dt} + x - 2z = 0, \\ \frac{dx}{dt} - \frac{dy}{dt} + \frac{dz}{dt} + x = 0, \\ \frac{dx}{dt} + \frac{dy}{dt} - \frac{dz}{dt} + x + 2y = 0. \end{cases}$$

解：原方程组可以化为 $\begin{cases} \frac{dx}{dt} = -x - y \\ \frac{dy}{dt} = -y - z \\ \frac{dz}{dt} = -z \end{cases} \Rightarrow A = \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{pmatrix} \Rightarrow$

$$|A - \lambda E| = \begin{vmatrix} -1-\lambda & -1 & 0 \\ 0 & -1-\lambda & -1 \\ 0 & 0 & -1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-1-\lambda)^3 = 0 \Rightarrow \lambda = -1 (\text{三重}).$$

$$\begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}^3 \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = 0$$

$$\Rightarrow v_1^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_2^{(1)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_3^{(1)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

$$v_1^{(2)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad v_2^{(2)} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}, \quad v_3^{(2)} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix},$$

$$v_1^{(3)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad v_2^{(3)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad v_3^{(3)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{-t}, \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix}_2 = \begin{pmatrix} -t \\ 1 \\ 0 \end{pmatrix} e^{-t}, \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix}_3 = \begin{pmatrix} \frac{t^2}{2} \\ -t \\ 1 \end{pmatrix} e^{-t}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -t \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} \frac{t^2}{2} \\ -t \\ 1 \end{pmatrix}) e^{-t}$$

$$14. \begin{cases} \frac{d^2x}{dt^2} = y, \\ \frac{d^2y}{dt^2} = x. \end{cases}$$

解： $\begin{cases} \frac{dx}{dt} = p \\ \frac{dp}{dt} = y \\ \frac{dy}{dt} = q \\ \frac{dq}{dt} = x \end{cases} \Rightarrow A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \Rightarrow |A - \lambda E| = \begin{vmatrix} -\lambda & 1 & 0 & 0 \\ 0 & -\lambda & 1 & 0 \\ 0 & 0 & -\lambda & 1 \\ 1 & 0 & 0 & -\lambda \end{vmatrix} =$

$$\lambda^4 - 1 = 0$$

$$\Rightarrow \lambda_1 = 1, \quad \lambda_2 = -1, \quad \lambda_3 = i, \quad \lambda_4 = -i$$

$$\Rightarrow v_1 = \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \\ \eta_1 \end{pmatrix} \Rightarrow \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \\ \eta_1 \end{pmatrix} = 0 \Rightarrow v_1 =$$

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix},$$

$$v_2 = \begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \\ \eta_2 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \\ \eta_2 \end{pmatrix} = 0 \Rightarrow v_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix},$$

$$v_3 = \begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \\ \eta_3 \end{pmatrix} \Rightarrow \begin{pmatrix} -i & 1 & 0 & 0 \\ 0 & -i & 1 & 0 \\ 0 & 0 & -i & 1 \\ 1 & 0 & 0 & -i \end{pmatrix} \begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \\ \eta_3 \end{pmatrix} = 0 \Rightarrow v_3 = \begin{pmatrix} 1 \\ i \\ -1 \\ -i \end{pmatrix}$$

$$\Rightarrow (\cos t + i \sin t) \begin{pmatrix} 1 \\ i \\ -1 \\ -i \end{pmatrix} \begin{pmatrix} \cos t + i \sin t \\ -\sin t + i \cos t \\ -\cos t - i \sin t \\ \sin t - i \cos t \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t} + c_3 \begin{pmatrix} \cos t \\ -\cos t \end{pmatrix} + c_4 \begin{pmatrix} \sin t \\ -\sin t \end{pmatrix}$$

$$15. \begin{cases} \frac{dx}{dt} = 2y - 5x + e^t, \\ \frac{dy}{dt} = x - 6y + e^{-2t}. \end{cases}$$

解：由第二个方程 $x = \frac{dy}{dt} + 6y - e^{-2t}$

$$\Rightarrow \frac{d^2 y}{dt^2} + 6 \frac{dy}{dt} + 2e^{-2t} = 2y - 5 \frac{dy}{dt} - 30y + 5e^{-2t} + e^t$$

$$\Rightarrow \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} + 28y = 3e^{-2t} + e^t.$$

$$\lambda^2 + 11\lambda + 28 = 0, \quad (\lambda + 4)(\lambda + 7) = 0, \quad \lambda_1 = -4, \quad \lambda_2 = -7$$

$$\Rightarrow \text{齐次方程的通解为 } \tilde{y} = c_1 e^{-4t} + c_2 e^{-7t},$$

$$y_0 = A e^{-2t} + B e^t, \quad y'_0 = -2A e^{-2t} + B e^t, \quad y''_0 = 4A e^{-2t} + B e^t.$$

$$\text{代入得 } A = \frac{3}{10}, \quad B = \frac{1}{40}$$

$$\Rightarrow y = c_1 e^{-4t} + c_2 e^{-7t} + \frac{3}{10} e^{-2t} + \frac{1}{40} e^t$$

$$\Rightarrow x = -4c_1 e^{-4t} - 7c_2 e^{-7t} - \frac{3}{5} e^{-2t} + \frac{1}{40} e^t \\ + 6c_1 e^{-4t} + 6c_2 e^{-7t} + \frac{3}{5} e^{-2t} + \frac{3}{20} e^t - e^{-2t}$$

$$= 2c_1 e^{-4t} - c_2 e^{-7t} + \frac{1}{5} e^{-2t} + \frac{7}{40} e^t$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2c_1 e^{-4t} - c_2 e^{-7t} + \frac{1}{5} e^{-2t} + \frac{7}{40} e^t \\ c_1 e^{-4t} + c_2 e^{-7t} + \frac{3}{10} e^{-2t} + \frac{1}{40} e^t \end{pmatrix}.$$

$$16. \begin{cases} \frac{dx}{dt} + \frac{dy}{dt} = -x + y + 3, \\ \frac{dx}{dt} - \frac{dy}{dt} = x + y - 3. \end{cases}$$

$$\text{解: } \begin{cases} \frac{dx}{dt} = y \\ \frac{dy}{dt} = -x + 3 \end{cases} \Rightarrow x'' = y' = -x + 3$$

$$\Rightarrow x = c_1 \cos t + c_2 \sin t + 3, \quad y = -c_1 \sin t + c_2 \cos t.$$

$$17. \begin{cases} \frac{dx}{dt} = 2x + 4y - e^{-t}, \\ \frac{dy}{dt} = -x + 2y - 4e^{-t}. \end{cases}$$

$$\text{解: } \lambda_{1,2} = 2 \pm 2i, \quad v_0 = \begin{pmatrix} -2i \\ 1 \end{pmatrix}$$

$$\Rightarrow x(t) = e^{2t} \begin{pmatrix} 2 \sin 2t & -2 \cos 2t \\ \cos 2t & \sin 2t \end{pmatrix}, \quad \text{特解为 } \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t} + c_1 e^{2t} \begin{pmatrix} 2 \sin 2t \\ \cos 2t \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} -2 \cos 2t \\ \sin 2t \end{pmatrix}.$$

$$18. \begin{cases} \frac{dx}{dt} - y = \cos t, \\ \frac{dy}{dt} + x = 1. \end{cases}$$

$$\text{解: } x'' = -x - \sin t + 1$$

$$\Rightarrow x = c_1 \cos t + c_2 \sin t + At \cos t + 1 \Rightarrow A = \frac{1}{2}$$

$$\Rightarrow x = c_1 \cos t + c_2 \sin t + \frac{t}{2} \cos t + 1$$

$$y = -c_1 \sin t + c_2 \cos t - \frac{t}{2} \sin t - \frac{\cos t}{2}.$$

$$19. \begin{cases} \frac{dx}{dt} + 5x + y = e^t, \\ \frac{dy}{dt} - x + 3y = e^{2t}. \end{cases}$$

$$\text{解: } A = \begin{pmatrix} -5 & 1 \\ 1 & -3 \end{pmatrix}, \quad \lambda_{1,2} = -4$$

$$\Rightarrow v_0^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_0^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad v_1^{(1)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad v_1^{(2)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \text{特}$$

$$\text{解为 } e^t \begin{pmatrix} \frac{4}{25} \\ \frac{1}{25} \end{pmatrix} + e^{2t} \begin{pmatrix} -\frac{1}{36} \\ \frac{1}{36} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = e^t \begin{pmatrix} \frac{4}{25} \\ \frac{1}{25} \end{pmatrix} + e^{2t} \begin{pmatrix} -\frac{1}{36} \\ \frac{7}{36} \end{pmatrix} + c_1 e^{-4t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-4t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

$$20. \begin{cases} \frac{dx}{dt} + \frac{dy}{dt} - x + 2y = 1 + e^t, \\ \frac{dy}{dt} + \frac{dz}{dt} + 2y + z = 2 + e^t, \\ \frac{dz}{dt} + \frac{dx}{dt} - x + z = 3 + e^t. \end{cases}$$

$$\text{解: } x' + y' + z' - x + 2y + z = 3 + \frac{3}{2}e^t$$

$$\Rightarrow x' = x + 1 + \frac{1}{2}e^t$$

$$\Rightarrow x = c_1 e^t - 1 + \frac{1}{2}te^t$$

$$\Rightarrow y' = -2y + \frac{1}{2}e^t$$

$$\Rightarrow y = c_2 e^{-2t} + \frac{1}{6}e^t$$

$$\Rightarrow z' = -z + 2 + \frac{1}{2}e^t$$

$$\Rightarrow z = c_3 e^{-t} + 2 + \frac{1}{4}e^t.$$

21. 试证明, 对于高阶线性方程 (3.9), 按第二章 §4 中二的变动任意常数法得到的通解, 与用变换 (3.3) 将 (3.9) 化成线性方程组 (3.9)' 之后, 按本节的变动任意常数法得到的通解 (3.42) 是一致的 (以你 n=2 情形证明之).

$$\text{解: } n=2, \quad \frac{d^2x}{dt^2} + P_1 \frac{dx}{dt} + P_2 x = f(t)$$

$$\Rightarrow \begin{cases} x'_1 = x_2 \\ x'_2 = -P_1 x_2 - P_2 x_1 - f \end{cases}$$

$$\Rightarrow A = \begin{pmatrix} 0 & 1 \\ -P_2 & -P_1 \end{pmatrix}, \quad \lambda^2 + P_1 \lambda + P_2.$$

易证相应的齐次方程的解是一致的, 只需证特解即可。

因为齐次方程解一致, 所以基本解矩阵与逆矩阵都一致, 特解也一致。得证。

22. 飞机在空中沿水平方向等速飞行, 速度为 v_0 , 一重为 mg 的炸弹从飞机上下落, 设空气的阻力为 R (常数), 试求炸弹运动规律。

$$\text{解: 设水平为 } x, \text{ 垂直为 } y, \text{ 则 } x(0) = y(0) = y'(0) = 0, \quad x'(0) = v_0$$

$$\Rightarrow mx'' = -R_x, \quad my'' = mg - R_y$$

$$\Rightarrow x = -\frac{R_x}{2} + c_1 t + c_2 \Rightarrow c_2 = 0, \quad c_1 = v_0$$

$$y = \frac{1}{2}(g - \frac{R_y}{m})t^2 + c_3 t + c_4 \Rightarrow c_3 = c_4 = 0$$

$$\Rightarrow x = -\frac{R_x}{2m}t^2 + v_0t, \quad y = \frac{1}{2}\left(g - \frac{R_y}{m}\right)t^2.$$

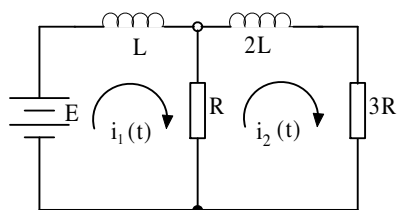


图 3-1

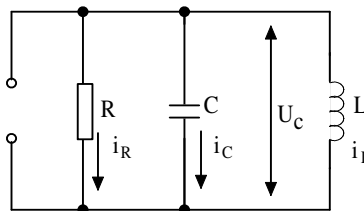


图 3-2

23. 设二电流回路如图 3-1, 电动势 E 为常数。若开始时电流 $i_1 = i_2 = 0$, 试求电流 $i_1(t)$, $i_2(t)$ 随时间 t 的变化规律。

$$\begin{aligned} \text{解: } & \begin{cases} 2Li_2' + 3Ri_2 = R(i_3 - i_2) \\ Li_1' + R(i_1 - i_2) = E \end{cases} \Rightarrow \lambda^2 + 3\lambda + \frac{3}{2} = 0 \\ \Rightarrow & \lambda_{1,2} = \frac{-3 \pm \sqrt{3}}{2}, \quad i_1(0) = 0, \quad i_2(0) = 0 \\ \Rightarrow & \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = e^{\frac{-3+\sqrt{3}}{2}t} \begin{pmatrix} -\frac{(2+\sqrt{3})E}{3R} \\ -\frac{(1+\sqrt{3})E}{6R} \end{pmatrix} + e^{\frac{-3-\sqrt{3}}{2}t} \begin{pmatrix} -\frac{(2-\sqrt{3})E}{3R} \\ -\frac{(1-\sqrt{3})E}{R} \end{pmatrix} + \begin{pmatrix} -\frac{4E}{3R} \\ \frac{E}{3R} \end{pmatrix}. \end{aligned}$$

24. 一电路如图 3-2 所示, 输入电压为零, 电路参数 $C = 1$ 法, $L = 1$ 亨, $R = 1$ 欧。试写出以电容上的电压 U_c 和电感上的电流 i_L 为未知函数, 以时间 t 为自变量的微分方程组。并设 $U_c(0) = U_{c0}$, $i_L(0) = i_{L0}$, 求方程组的特解。

$$\begin{aligned} \text{解: } & \begin{cases} Li_L' = -U_c \\ cU_c' = i_c = i_L - i_R = i_L - \frac{U_c}{R} \end{cases} \\ \Rightarrow & \begin{cases} Li_L' = -U_c \\ U_c' = i_L - U_c \end{cases} \\ U_c(0) &= U_{c0}, \quad i_L(0) = i_{L0} \\ \Rightarrow & \lambda^2 + \lambda + 1 = 0, \quad \lambda_{1,2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i \\ \Rightarrow & U_c(t) = U_{c0}e^{-\frac{1}{2}t} \cos \frac{\sqrt{3}}{2}t - \frac{2i_{L0} + U_{c0}}{\sqrt{3}}e^{-\frac{1}{2}t} \sin \frac{\sqrt{3}}{2}t \\ i_L(t) &= \frac{1}{2}U_{c0}e^{-\frac{1}{2}t}(-\cos \frac{\sqrt{3}}{2}t + \sqrt{3} \sin \frac{\sqrt{3}}{2}t) + \frac{2i_{L0} + U_{c0}}{2\sqrt{3}}e^{-\frac{1}{2}t}(\sqrt{3} \cos \frac{\sqrt{3}}{2}t + \sin \frac{\sqrt{3}}{2}t). \end{aligned}$$

25. 质量为 m_1 和 m_2 的两个小球，穿在一光滑水平杆上，由一轻质弹簧连接，且可沿杆移动。当弹簧不受力时，两小球重心间的距离为 l 。若用 x_1 ， x_2 分别表示两小球的位移，并设 $x_1(0) = 0$ ， $\dot{x}_1(0) = v_0$ ， $x_2(0) = l$ ， $\dot{x}_2(0) = 0$ 。试求两球的运动规律（这里记号 $\cdot$ 表示 $\frac{d}{dt}$ ）。

$$\text{解: } \begin{cases} m_1 x_1'' = -k[l - (x_2 - x_1)] \\ m_2 x_2'' = k[l - (x_2 - x_1)] \end{cases}$$

$$x_1(0) = 0, \quad x_2(0) = l, \quad \dot{x}_1 = v_0, \quad \dot{x}_2(0) = 0.$$

添加未知元 y_1 ， y_2 ，其中 $y_1 = x_1'$ ， $y_2 = x_2'$

$$\Rightarrow \begin{cases} m_1 y_1' = -k[l - (x_2 - x_1)] \\ x_1' = y_1 \\ m_2 y_2' = k[l - (x_2 - x_1)] \\ x_2' = y_2 \end{cases} \quad \begin{cases} x(0) = 0 \\ x_2(0) = l \\ y_1(0) = v_0 \\ y_2(0) = 0 \end{cases}$$

$$\Rightarrow \lambda_{1,2} = 0, \quad \lambda_{3,4} = \pm i \sqrt{\frac{k}{m_1} + \frac{k}{m_2}}.$$

$$\text{令 } w^2 = \frac{k}{m_1 m_2} (m_1 + m_2)$$

$$\Rightarrow \begin{aligned} x_1 &= \frac{m_1 m_2}{m_1 + m_2} \left(m_1 t + \frac{v_0}{w} \sin wt \right) \\ x_2 &= \frac{m_1 m_2}{m_1 + m_2} \left(m_1 t - \frac{v_0}{w} \sin wt \right) + l. \end{aligned}$$